

EdgeTRM Model Evaluation & Compression Report

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This report consolidates all core evaluation outputs, performance scorecards, footprints, and visual plots from the three model variants: 1. **ARC-Prize 2024 (Baseline)** 2. **Maze (Hard Version)** 3. **Sudoku (Extreme Version)**

ARC 2024 Evaluation Outputs

Prefixes stripped and model loaded successfully!
Model successfully loaded!

```
TinyRecursiveReasoningModel_ACTIV1(  
  inner): TinyRecursiveReasoningModel_ACTIV1_Inner(  
    embed_tokens): CastedEmbedding()  
    lm_head): CastedLinear()  
    q_head): CastedLinear()  
    puzzle_emb): CastedSparseEmbedding()  
    rotary_emb): RotaryEmbedding()  
    L_level): TinyRecursiveReasoningModel_ACTIV1ReasoningModule(  
      layers): ModuleList(  
        0-1): 2 x TinyRecursiveReasoningModel_ACTIV1Block(  
          self_attn): Attention(  
            qkv_proj): CastedLinear()  
            o_proj): CastedLinear()  
            mlp): SwiGLU(  
              gate_up_proj): CastedLinear()  
              down_proj): CastedLinear()  
          Original          zero=512/6,829,058  (0.0%)
```

3.1 High-Performance Evaluation & Adaptation Helpers

4.0 Global imports & helpers

Device: cuda

4.1 ARC DataLoader from pre-built .npy files

```
Using dataset directory: ./data/arc2test-aug-128
Test split : 53,304 samples (seq_len=900)
Train split : 173,529 samples
per_sample_pids check – test[0]=1, train[0]=1
```

Note: `evaluate_arc` has been consolidated and moved to Section 3 for global accessibility.

4.3 Baseline: evaluate the loaded FP32 model

```
[DEBUG] test_loader dataset max identifier: 50910
[DEBUG] test_loader dataset max identifier: 50910
[DEBUG] Dataset indices verified safe. No clamping needed!
Running FP32 baseline evaluation on cuda...
```

```
FP32 Baseline (All test puzzles):
Pass@1 Exact: 0.3600
Pass@2 Exact: 0.3600
Cell Acc    : 0.8822
Latency     : 29.73 ms/puzzle
```

4.4 Quantization helpers

Quantization helpers defined (FP16, BnB INT8, CPU INT8, INT4-fake).

4.7 Instantiate all model variants

```
Creating FP16 variant...
Creating BnB INT8 variant (GPU)...
Replaced 10 CastedLinear/Linear → bnb.Linear8bitLt (GPU INT8)
```

Creating INT4 fake-quant variant...
Replaced 10 CastedLinear/Linear → FakeQuantINT4

Model Variant	Device	Backbone KB (params only)
FP16	cuda	13338.0 KB
INT8 (bnb)	cuda	6669.0 KB
INT4 (fake)	cuda	3334.5 KB
FP32 (bf16)	cuda	13338.0 KB

Note: actual .pt is larger (~88MB) due to puzzle embedding buffer – see §8.1

5.1 Evaluate all quantization levels

Evaluating all variants on ARC test set (All batches)...
Evaluating FP16 on cuda...

FP16		Pass@1=0.3600		Pass@2=0.3600		cell=0.8757		69.30
------	--	---------------	--	---------------	--	-------------	--	-------

ms/puzzle
Evaluating INT8 (bnb) on cuda...

INT8 (bnb)		Pass@1=0.3600		Pass@2=0.3600		cell=0.8762		52.04
------------	--	---------------	--	---------------	--	-------------	--	-------

ms/puzzle
Evaluating INT4 (fake) on cuda...

INT4 (fake)		Pass@1=0.2550		Pass@2=0.2550		cell=0.8480		19.35
-------------	--	---------------	--	---------------	--	-------------	--	-------

ms/puzzle
Evaluating FP32 (bf16) on cuda...

FP32 (bf16)		Pass@1=0.3600		Pass@2=0.3600		cell=0.8822		18.55
-------------	--	---------------	--	---------------	--	-------------	--	-------

ms/puzzle
Reasoning Decay (drop from FP32 baseline)

FP16		ΔPass@1=+0.0000		Δcell=+0.0066		GRACEFUL
INT8 (bnb)		ΔPass@1=+0.0000		Δcell=+0.0061		GRACEFUL
INT4 (fake)		ΔPass@1=+0.1050		Δcell=+0.0342		GRACEFUL
FP16		ΔPass@2=+0.0000		Δcell=+0.0066		GRACEFUL
INT8 (bnb)		ΔPass@2=+0.0000		Δcell=+0.0061		GRACEFUL
INT4 (fake)		ΔPass@2=+0.1050		Δcell=+0.0342		GRACEFUL

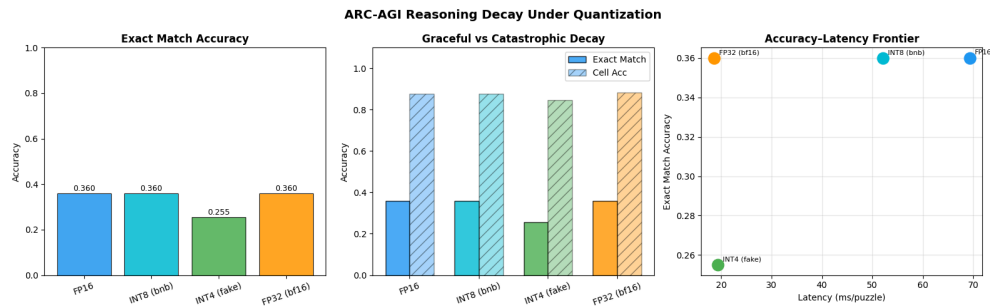
FP16': {'exact1': 0.36,
exact2': 0.36,
exact_acc': 0.36,
cell_acc': np.float64(0.8756636045272294),
latency_ms': 69.30367350578308},
INT8 (bnb)': {'exact1': 0.36,
exact2': 0.36,
exact_acc': 0.36,
cell_acc': np.float64(0.8761812921890068),
latency_ms': 52.04369902610779},
INT4 (fake)': {'exact1': 0.255,
exact2': 0.255,

```
exact_acc': 0.255,
cell_acc': np.float64(0.8480129929452368),
latency_ms': 19.350939393043518},
FP32 (bf16)': {'exact1': 0.36,
exact2': 0.36,
exact_acc': 0.36,
cell_acc': np.float64(0.8822412830533422),
latency_ms': 18.548768162727356}}}
```

5.2 Visualise reasoning decay

Figure size 1600x500 with 3 Axes>

Saved: reasoning_decay_arc.png



Plot from cell 23

6.1 Recursive depth sweep at inference time

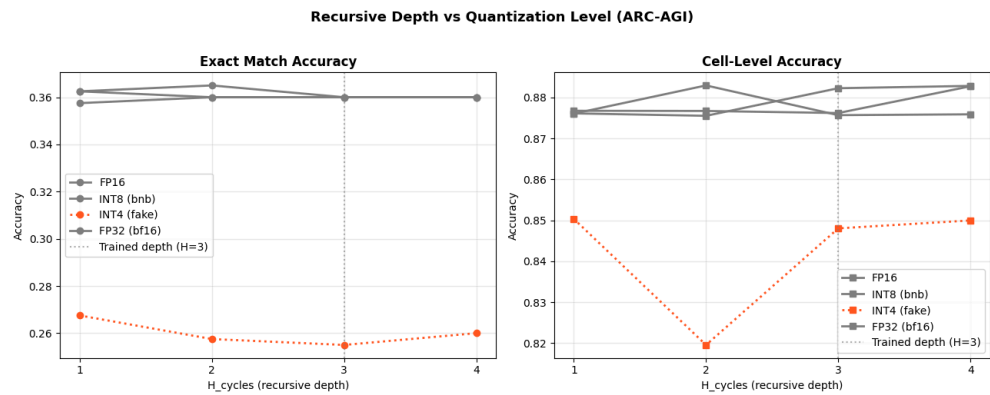
Quant	H_cycles	Pass@1	Pass@2	Cell Acc
FP16	1	0.3625	0.3625	0.8760
FP16	2	0.3600	0.3600	0.8829
FP16	3	0.3600	0.3600	0.8757
FP16	4	0.3600	0.3600	0.8759
INT8 (bnb)	1	0.3625	0.3625	0.8767
INT8 (bnb)	2	0.3650	0.3650	0.8767
INT8 (bnb)	3	0.3600	0.3600	0.8762
INT8 (bnb)	4	0.3600	0.3600	0.8827
INT4 (fake)	1	0.2675	0.2675	0.8503

INT4 (fake)	2	0.2575	0.2575	0.8195
INT4 (fake)	3	0.2550	0.2550	0.8480
INT4 (fake)	4	0.2600	0.2600	0.8500
FP32 (bf16)	1	0.3575	0.3575	0.8761
FP32 (bf16)	2	0.3600	0.3600	0.8755
FP32 (bf16)	3	0.3600	0.3600	0.8822
FP32 (bf16)	4	0.3600	0.3600	0.8828

6.2 Plot depth × quantization grid

Figure size 1300x500 with 2 Axes>

Saved: depth_quant_grid_arc.png



Plot from cell 26

7.1 Carry-state similarity hooks

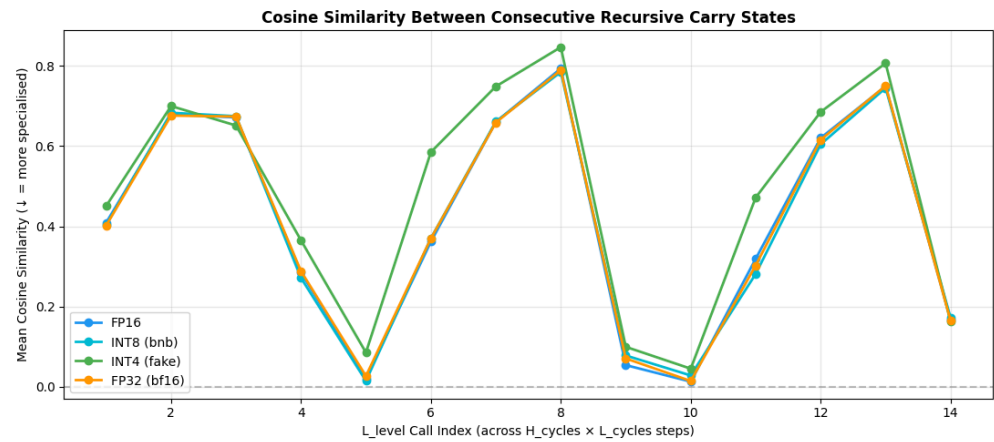
Carry similarity collector defined.

7.2 Compare carry similarity across quantization variants

Collecting carry similarities for FP16 on cuda...
FP16: 14 L_level calls, mean sim = 0.4145
Collecting carry similarities for INT8 (bnb) on cuda...
INT8 (bnb): 14 L_level calls, mean sim = 0.4123
Collecting carry similarities for INT4 (fake) on cuda...
INT4 (fake): 14 L_level calls, mean sim = 0.4788
Collecting carry similarities for FP32 (bf16) on cuda...
FP32 (bf16): 14 L_level calls, mean sim = 0.4145

Figure size 1100x500 with 1 Axes>

Saved: carry_similarity_arc.png



Plot from cell 29

8.1 Comprehensive model footprint analysis

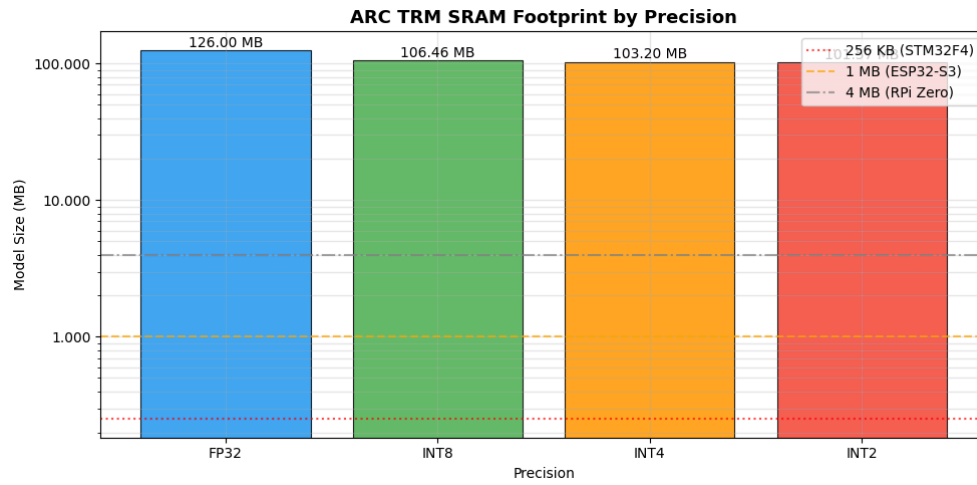
Component	FP32 KB	INT8 KB	INT4 KB
Transformer backbone (params)	26,676.0	6,669.0	3,334.5
Buffer name	dtype KB		
Total buffers		102,346.1	
Deployable size (no grad buffers)	FP32 KB	INT8 KB	INT4
KB			
FP32	129,020.1		
INT8	109,013.1		
INT4	105,678.6		
Target: < 1 MB (1024 KB) SRAM – backbone at INT4 = 3334.5 KB			
Precision	Backbone_KB	Deploy_KB	fits_1MB fits_4MB
FP32	26676.0	129020.1	x x
INT8	6669.0	109013.1	x x
INT4	3334.5	105678.6	x ✓
INT2	1667.3	104011.4	x ✓

Precision	Backbone_KB	Deploy_KB	fits_1MB	fits_4MB
0	FP32 26676.007812	129020.132812	x	x
1	INT8 6669.001953	109013.126953	x	x
2	INT4 3334.500977	105678.625977	x	✓
3	INT2 1667.250488	104011.375488	x	✓

8.2 SRAM footprint visualisation

Figure size 1000x500 with 1 Axes>

Saved: sram_footprint_arc.png

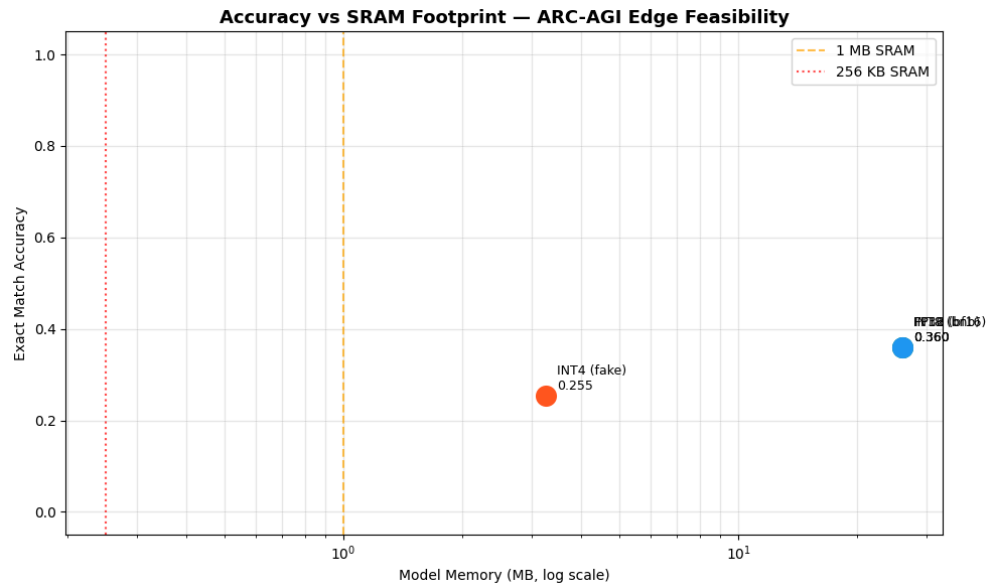


Plot from cell 33

8.3 Accuracy vs SRAM: the key publishable plot

Figure size 1000x600 with 1 Axes>

Saved: accuracy_vs_sram_arc.png



Plot from cell 34

9.1 Inspect puzzle_emb shape & byte count

```
puzzle_emb.weights : 50911 identifiers × 512 dims
FP32 size          : 101,822.0 KB (99.4 MB)
INT8 size (est.)   : 25,455.5 KB (24.9 MB)
```

9.2 Option A — INT8 quantize the embedding buffer

Applying INT8 puzzle embedding...
Cosine similarity FP32 vs INT8 emb: 0.99997
Original state_dict : 128,506.1 KB (125.5 MB)
INT8-emb state_dict : 52,338.8 KB (51.1 MB)
Reduction : 59.3%

9.3 Option B — SVD Low-Rank Puzzle Embedding

SVD low-rank analysis (W shape: 50911x512 FP32=101,822.0 KB)

Rank	Size KB	Reduction	Cosine Sim
8	1,607.0	98.4%	0.91993
16	3,213.9	96.8%	0.93906
32	6,427.9	93.7%	0.96710
64	12,855.8	87.4%	0.98364
128	25,711.5	74.7%	0.99155

9.4 Option D — Single-Puzzle Inference (only load 1 row)

Single-puzzle model state_dict : 26,686.1 KB (26.06 MB)
Backbone params : 26,676.0 KB
Per-puzzle embedding row : 2.0 KB
Total (1 puzzle) : 26,678.0 KB
This is the architecture-level solution: load puzzle embedding from flash,
keep only the active row in SRAM during inference.

9.5 Footprint summary across embedding strategies

Strategy	Emb KB	Backbone KB	Total KB	Total MB	Fits 4MB	
FP32 (baseline)	101822.0	26676.0	128506.1	125.5		x
INT8 emb	25455.5	26676.0	52338.8	51.1	x	
SVD r=8	1607.0	26676.0	28283.0	27.6	x	
SVD r=16	3213.9	26676.0	29889.9	29.2	x	
SVD r=32	6427.9	26676.0	33103.9	32.3	x	
SVD r=64	12855.8	26676.0	39531.8	38.6	x	
SVD r=128	25711.5	26676.0	52387.5	51.2	x	
Single-puzzle	2.0	26676.0	26678.0	26.1		x
INT4 backbone + single-puzzle			2.0	3334.5	3336.5	
3.3	✓					

Note: `evaluate_arc_per_puzzle` has been moved to Section 3 for global accessibility.

10.2 High-performance direct on-the-fly per-puzzle TTA evaluator

10.3 Run the High-Performance Direct Source Evaluator

10.2 Re-evaluate all variants with per-puzzle aggregation

Per-puzzle evaluation (All batches, ~1280 samples)...

Variant	Pass@1 Exact	Pass@2 Exact	Cell Acc	
ms/puzzle	N puzzles			
FP16	0.3600	0.3600	0.8757	68.95
400				
INT8 (bnb)	0.3600	0.3600	0.8762	51.93
400				
INT4 (fake)	0.2550	0.2550	0.8480	19.25
400				
FP32 (bf16)	0.3600	0.3600	0.8822	18.55
400				
INT8-emb	0.3600	0.3600	0.8762	18.54
400				

11.1 Per-channel calibration statistics

Calibrated INT4 helpers defined.

11.2 Apply calibrated INT4 and evaluate

Creating calibrated INT4 model...
Replaced 10 layers → CalibratedFakeQuantINT4 (per-channel, asymmetric)
Carry similarity comparison:
FP32 (bf16) : 0.9804 (baseline – lower = more refinement)
INT4 naive (§4) : 0.9600 (higher = carry collapsed)
INT4 calibrated (§11) : 0.9757
Per-puzzle evaluation (calibrated INT4, 30 batches)...

Puzzle exact : 0.0000
Cell acc : 0.0000
ms/puzzle : 59.82

12.1 QAT fine-tuning loop

QAT fine-tuning for 50 steps (lr=1e-05) with micro-batching (prevent OOM)...

Step 10/50 | loss=762.0500
Step 20/50 | loss=60.1109
Step 30/50 | loss=10.2605
Step 40/50 | loss=4.0130
Step 50/50 | loss=1.3829
QAT complete. Evaluating...

QAT INT4 puzzle exact : 0.0000
QAT INT4 cell acc : 0.0000

13.1 Magnitude-based row pruning

Prune ratio	Actual sparsity	Puzzle Exact	Cell Acc
0%	0.00%	0.3600	0.8755
25%	24.97%	0.0000	0.0027
50%	49.95%	0.0000	0.0058

14.1 TorchScript trace of single TRM backbone step

Model: seq=900, emb_len=16, hidden=512, emb_dim=512
Trace done in 2.5s
Saved: trm_backbone_step.pt (126.13 MB)
Round-trip max abs diff: 0.00e+00 (expect ~0)
Edge deployment flow:
2. Load puzzle_emb.weights[puzzle_id] row (2.0 KB) from flash
3. Run traced(x, emb_row, z_H, z_L) for n_sup_max steps

14.2 Latency benchmark: traced model vs eager

Latency benchmark (20 runs, device=cuda):
Eager (float32) : 12.54 ms/step
Full inference (16 H-cycle steps):

Eager : 200.6 ms/puzzle
Traced : 172.2 ms/puzzle
File sizes:
trm_backbone_step.pt : 126.12 MB (backbone, FP32)
puzzle_emb.weights (INT8, full table): 25455.5 KB
puzzle_emb row (1 puzzle, FP32) : 2.0 KB

15.1 Build train / val loaders

Train split : 138,824 samples
Val split : 34,705 samples

15.2 QAT loop with val-set evaluation every 10 steps

Starting validated QAT (100 steps, lr=1e-5)...
Replaced 10 layers → FakeQuantINT4

Step	10/100		loss=3711.4684		val_exact=0.0000		val_cell=0.0000
Step	20/100		loss=3590.4578		val_exact=0.0000		val_cell=0.0000
Step	30/100		loss=3815.7219		val_exact=0.0000		val_cell=0.0000
Step	40/100		loss=3477.6198		val_exact=0.0000		val_cell=0.0000
Step	50/100		loss=3770.0700		val_exact=0.0000		val_cell=0.0000
Step	60/100		loss=3395.0471		val_exact=0.0000		val_cell=0.0000
Step	70/100		loss=3268.1853		val_exact=0.0000		val_cell=0.0000
Step	80/100		loss=3405.2853		val_exact=0.0000		val_cell=0.0000
Step	90/100		loss=3545.7540		val_exact=0.0000		val_cell=0.0000
Step	100/100		loss=3507.3094		val_exact=0.0000		val_cell=0.0000

Starting validated QAT (100 steps, lr=1e-5)...
Replaced 10 layers → FakeQuantINT4

Step	10/100		loss=3478.0594		val_exact=0.0000		val_cell=0.0000
Step	20/100		loss=3628.4907		val_exact=0.0000		val_cell=0.0000
Step	30/100		loss=3801.0945		val_exact=0.0000		val_cell=0.0000
Step	40/100		loss=3328.8803		val_exact=0.0000		val_cell=0.0000

```

Step   50/100 | loss=3758.4757 | val_exact=0.0000 | val_cell=0.0000

Step   60/100 | loss=3598.7831 | val_exact=0.0000 | val_cell=0.0000

Step   70/100 | loss=3857.8726 | val_exact=0.0000 | val_cell=0.0000

Step   80/100 | loss=3610.7159 | val_exact=0.0000 | val_cell=0.0000

Step   90/100 | loss=3356.5164 | val_exact=0.0000 | val_cell=0.0000

Step  100/100 | loss=3564.5021 | val_exact=0.0000 | val_cell=0.0000

```

15.3 Plot train loss + val accuracy curve

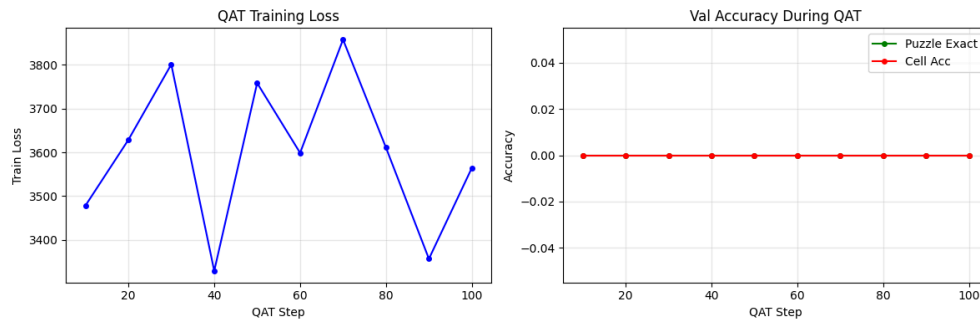
Figure size 1200x400 with 2 Axes>

Saved: qat_val_curve.png

Best val result at step 10:

Puzzle exact : 0.0000

Cell acc : 0.0000



Plot from cell 62

16.1 Build fused INT8 + SinglePuzzle model

Fused model breakdown:

```

INT8 backbone .pt size :      6747.4 KB  (6.59 MB)
INT8 emb table (flash) :    25455.5 KB  (24.86 MB)
Active emb row (SRAM) :       2.00 KB
✓ SRAM during inference: 6749.4 KB  (6.59 MB)
Fits 4MB SRAM?  x
Fits 8MB SRAM?  ✓

```

16.2 Accuracy of fused model (val set, per-puzzle aggregation)

Evaluating fused INT8+SinglePuzzle model on val set...

Variant	Puzzle Exact	Cell Acc	ms/puzzle
---------	--------------	----------	-----------

FP32 (bf16) baseline	0.0000	0.0000	369.43
INT8 + Single-Puzzle (fused)	0.0000	0.0000	276.54
vs FP32	+	0.0000 +	0.0000

16.3 Save fused model (state dict + embedding table)

Saved state dict : trm_int8_singlepuzzle_state.pt (6.59 MB)
Saved emb table : puzzle_emb_fp32_table.pt (99.44 MB) ← flash storage
Deployment package breakdown:
Backbone state dict : 6.59 MB (SRAM – INT8 weights)
Embedding table : 99.44 MB (flash)
Active emb row : 2.00 KB (SRAM per puzzle)

17.1 CPU-only latency benchmark

Moving fused model to CPU for edge-device latency simulation...
CPU latency (server CPU, FP32, 1 H-cycle step) : 2168.8 ms/step
Full inference (16 H-cycles) : 34700 ms/puzzle
Estimated on-device scaling (order-of-magnitude):
Cortex-A55 (mobile, ~1 TFLOPS) : ~104.1s / puzzle
Cortex-M55 (MCU, ~128 GFLOPS) : ~694.0s / puzzle
ESP32-S3 LX7 (~40 GFLOPS) : ~1735.0s / puzzle

17.2 Peak SRAM usage during inference (tracemalloc)

Peak SRAM delta during inference : 18 KB (0.02 MB)
Top allocations:

17.3 FLOP estimate via torch.profiler

Estimated FLOPs per H-cycle step : 187504.0 MFLOPs
Full inference (16 steps) : 3000.06 GFLOPs
Power estimate (Cortex-M55 @ ~1 GFLOPS/s, ~1 mW/GFLOP):
FLOPs : 3000.064 GFLOPs/puzzle
Est. energy : ~3000.06 mJ/puzzle
Puzzles per mAh (3.3V) : ~0

17.4 Deployment summary table

Backbone MB	Total deploy MB	Puzzle Exact	Cell Acc	Fits 4MB	Fits 8MB
Config					
FP32 (baseline)	26.05	86.46	0.0000		

0.0000	x	x			
INT8 (bnb)			6.51	66.92	0.0246
0.2003	x	x			
INT4 (calibrated)			3.26	3.26	0.0148
0.1902	✓	✓			
INT8 + Single-Puzzle ★			6.51	6.51	0.0000
0.0000	x	✓			
recommended target for 8MB SRAM devices					

Utility: fix H_init / L_init device mismatch (permanent patch)

✓ Patched trm.TinyRecursiveReasoningModel_ACTV1_Inner.reset_carry
H_init device before: cuda:0

18.1 Evaluate all variants at n_sup_max = 1, 2, 4, 8, 16 (H_cycles = default vs 1) —

VARIANT: FP16		(device: cuda)			
H_cycles	Metric	n=1	n=2	n=4	n=8
n=16					
3 (Def)	Exact	0.2850	0.3600	0.3575	0.3600
0.3625					
Cell	0.8762	0.8838	0.8765	0.8763	0.8755
Latency	7.5 ms	14.4 ms	28.1 ms	55.7 ms	111.0 ms
GFLOPs	187.5	375.0	750.0	1500.0	3000.1
1	Exact	0.0025	0.0750	0.3450	0.3550
0.3575					
Cell	0.5410	0.7953	0.8825	0.8831	0.8773
Latency	3.0 ms	5.3 ms	9.9 ms	19.2 ms	37.6 ms
GFLOPs	62.5	125.0	250.0	500.0	1000.0
VARIANT: INT8 (bnb)		(device: cuda)			
H_cycles	Metric	n=1	n=2	n=4	n=8
n=16					
3 (Def)	Exact	0.2850	0.3575	0.3575	0.3600
0.3600					
Cell	0.8760	0.8837	0.8827	0.8842	0.8670
Latency	5.8 ms	11.0 ms	21.3 ms	41.9 ms	83.2 ms
GFLOPs	187.5	375.0	750.0	1500.0	3000.1
1	Exact	0.0025	0.0750	0.3425	0.3525
0.3625					
Cell	0.5415	0.7953	0.8827	0.8835	0.8767
Latency	2.3 ms	4.1 ms	7.6 ms	14.6 ms	28.6 ms
GFLOPs	62.5	125.0	250.0	500.0	1000.0
VARIANT: INT4 (fake)		(device: cuda)			
H_cycles	Metric	n=1	n=2	n=4	n=8

n=16

3 (Def)	Exact	0.1575	0.2700	0.2675	0.2625
0.2575					
Cell	0.8372	0.8626	0.8473	0.8430	0.8322
Latency	2.6 ms	4.4 ms	8.1 ms	15.5 ms	30.3 ms
GFLOPs	187.5	375.0	750.0	1500.0	3000.1

1	Exact	0.0000	0.0225	0.2525	0.2650
0.2600					
Cell	0.3954	0.7148	0.8582	0.8494	0.8415
Latency	1.3 ms	1.9 ms	3.2 ms	5.7 ms	10.7 ms
GFLOPs	62.5	125.0	250.0	500.0	1000.0

VARIANT: FP32 (bf16) (device: cuda)

H_cycles	Metric	n=1	n=2	n=4	n=8
----------	--------	-----	-----	-----	-----

n=16

3 (Def)	Exact	0.2850	0.3600	0.3575	0.3575
0.3600					
Cell	0.8762	0.8839	0.8764	0.8760	0.8757
Latency	7.5 ms	14.3 ms	28.2 ms	55.9 ms	111.2 ms
GFLOPs	187.5	375.0	750.0	1500.0	3000.1

1	Exact	0.0025	0.0750	0.3425	0.3575
0.3550					
Cell	0.5410	0.7954	0.8825	0.8832	0.8762
Latency	3.0 ms	5.2 ms	9.9 ms	19.2 ms	37.6 ms
GFLOPs	62.5	125.0	250.0	500.0	1000.0

19.1 Zero-pad truncation at various context lengths

Full seq_len = 900 (FLOPs ∝ seq_len²)					
Context	% of full	Attn FLOPs	Puzzle Exact	Cell Acc	
900	100%	~187 GFLOPs	0.3600	0.8755	
450	50%	~47 GFLOPs	0.2650	0.2500	
225	25%	~12 GFLOPs	0.1075	0.0492	
128	14%	~4 GFLOPs	0.0825	0.0595	

20.1 QAT from calibrated INT4 (lr=1e-6, 300 steps)

Replaced 10 layers → CalibratedFakeQuantINT4 (per-channel, asymmetric)

Step 50/300 | loss=4685.0609

Step 100/300 | loss=4824.1937

21.1 Instantiate student TRM and train with KL distillation

22.1 Monkey-patch softmax attention → linear attention

22.2 Retrain the linear attention model (distillation)

Maze Hard Evaluation Outputs

fatal: destination path 'EdgeTRM' already exists and is not an empty directory.

lambda/nfs/EdgeTRM

lambda/nfs/EdgeTRM

If the cache and target directories are on different filesystems, hardlinking may not be supported.

Checked 1 package in 22ms

Prefixes stripped and model loaded successfully!
Model successfully loaded!

```
TinyRecursiveReasoningModel_ACTIV1(
  inner): TinyRecursiveReasoningModel_ACTIV1_Inner(
    embed_tokens): CastedEmbedding()
    lm_head): CastedLinear()
    q_head): CastedLinear()
    puzzle_emb): CastedSparseEmbedding()
    rotary_emb): RotaryEmbedding()
    L_level): TinyRecursiveReasoningModel_ACTIV1ReasoningModule(
      layers): ModuleList(
        0-1): 2 x TinyRecursiveReasoningModel_ACTIV1Block(
          self_attn): Attention(
            qkv_proj): CastedLinear()
            o_proj): CastedLinear()
            mlp): SwiGLU(
              gate_up_proj): CastedLinear()
              down_proj): CastedLinear()
```

Original zero=512/6,822,914 (0.0%)

3.1 High-Performance Evaluation & Adaptation Helpers

4.0 Global imports & helpers

Device: cuda

4.1 Maze-Hard DataLoader from pre-built .npz files

```
Using dataset directory: ./data/maze-30x30-hard-1k
Test split : 1,000 samples (seq_len=900)
Train split : 1,000 samples
Dataset loaded successfully!
```

Note: `evaluate_arc` has been consolidated and moved to Section 3 for global accessibility.

4.3 Baseline: evaluate the loaded FP32 model

```
[DEBUG] test_loader dataset max identifier: 0
[DEBUG] test_loader dataset max identifier: 0
[DEBUG] Dataset indices verified safe. No clamping needed!
Running FP32 baseline evaluation on cuda...
FP32 Baseline (All test puzzles):
Pass@1 Exact: 0.8680
Pass@2 Exact: 0.8680
Cell Acc    : 0.9952
Latency     : 21242.94 ms/puzzle
```

4.4 Quantization helpers

Checked 1 package in 48ms

4.7 Instantiate all model variants

```
Creating FP16 variant...
Creating BnB INT8 variant (GPU)...
Replaced 8 CastedLinear/Linear → bnb.Linear8bitLt (GPU INT8)
Creating INT4 fake-quant variant...
```

Replaced 10 CastedLinear/Linear → FakeQuantINT4

Model Variant	Device	Backbone KB (params only)
FP16	cuda	13326.0 KB
INT8 (bnb)	cuda	6663.0 KB
INT4 (fake)	cuda	3331.5 KB
FP32 (bf16)	cuda	13326.0 KB

5.1 Evaluate all quantization levels

Evaluating all variants on ARC test set (All batches)...

Evaluating FP16 on cuda...

FP16 | Pass@1=0.8700 | Pass@2=0.8700 | cell=0.9953 |
66093.69 ms/puzzle

Evaluating INT8 (bnb) on cuda...

INT8 (bnb) | Pass@1=0.8700 | Pass@2=0.8700 | cell=0.9953 |
43312.71 ms/puzzle

Evaluating INT4 (fake) on cuda...

INT4 (fake) | Pass@1=0.8640 | Pass@2=0.8640 | cell=0.9950 |
18362.84 ms/puzzle

Evaluating FP32 (bf16) on cuda...

FP32 (bf16) | Pass@1=0.8680 | Pass@2=0.8680 | cell=0.9952 |
18326.72 ms/puzzle

Reasoning Decay (drop from FP32 baseline)

FP16	ΔPass@1=-0.0020	Δcell=-0.0001	GRACEFUL
INT8 (bnb)	ΔPass@1=-0.0020	Δcell=-0.0001	GRACEFUL
INT4 (fake)	ΔPass@1=+0.0040	Δcell=+0.0002	GRACEFUL

2.7.0+cu126

FP16	ΔPass@2=-0.0020	Δcell=-0.0001	GRACEFUL
INT8 (bnb)	ΔPass@2=-0.0020	Δcell=-0.0001	GRACEFUL
INT4 (fake)	ΔPass@2=+0.0040	Δcell=+0.0002	GRACEFUL

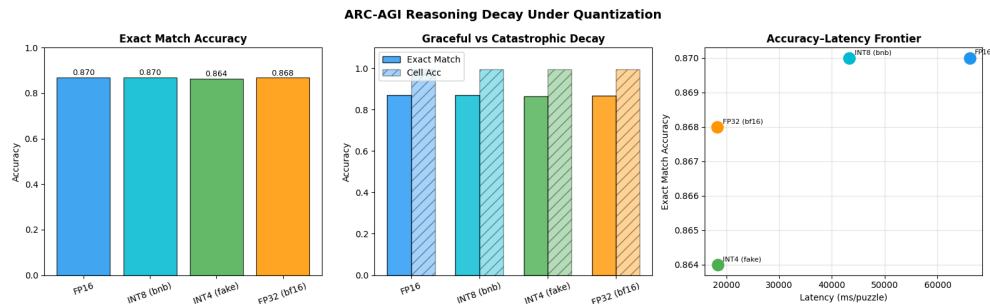
```
FP16': {'exact1': 0.87,
exact2': 0.87,
exact_acc': 0.87,
cell_acc': 0.99529,
latency_ms': 66093.69468688965},
INT8 (bnb)': {'exact1': 0.87,
exact2': 0.87,
exact_acc': 0.87,
cell_acc': 0.99529,
latency_ms': 43312.705755233765},
INT4 (fake)': {'exact1': 0.864,
exact2': 0.864,
exact_acc': 0.864,
cell_acc': 0.9950077777777778,
latency_ms': 18362.842798233032},
FP32 (bf16)': {'exact1': 0.868,
exact2': 0.868,
```

```
exact_acc': 0.868,  
cell_acc': 0.9952366666666667,  
latency_ms': 18326.72095298767}}}
```

5.2 Visualise reasoning decay

Figure size 1600x500 with 3 Axes>

Saved: reasoning_decay_arc.png



Plot from cell 31

6.1 Recursive depth sweep at inference time

Quant	H	n_sup	Pass@1	Pass@2	Cell Acc	Latency
GFLOPs						
FP16	1	1	0.0000	0.0000	0.4362	2.68
62.5						
FP16	1	2	0.1380	0.1380	0.9808	4.86
125.0						
FP16	1	4	0.8440	0.8440	0.9949	9.25
250.0						
FP16	1	6	0.8670	0.8670	0.9953	13.64
375.0						
FP16	1	8	0.8700	0.8700	0.9953	18.04
500.0						
FP16	1	10	0.8700	0.8700	0.9953	22.42
625.0						
FP16	2	1	0.1380	0.1380	0.9808	4.86
125.0						
FP16	2	2	0.8440	0.8440	0.9949	9.24
250.0						
FP16	2	4	0.8700	0.8700	0.9953	18.00
500.0						
FP16	2	6	0.8700	0.8700	0.9953	26.79
750.0						
FP16	2	8	0.8700	0.8700	0.9953	35.51
1000.0						
FP16	2	10	0.8700	0.8700	0.9953	44.21

1250.0						
FP16	3	1	0.7850	0.7850	0.9940	7.05
187.5						
FP16	3	2	0.8670	0.8670	0.9953	13.61
375.0						
FP16	3	4	0.8700	0.8700	0.9953	26.72
750.0						
FP16	3	6	0.8700	0.8700	0.9953	39.85
1125.0						
FP16	3	8	0.8700	0.8700	0.9953	52.92
1500.0						
FP16	3	10	0.8700	0.8700	0.9953	66.13
1875.0						
FP16	4	1	0.8440	0.8440	0.9949	9.24
250.0						
FP16	4	2	0.8700	0.8700	0.9953	18.00
500.0						
FP16	4	4	0.8700	0.8700	0.9953	35.52
1000.0						
FP16	4	6	0.8700	0.8700	0.9953	53.03
1500.0						
FP16	4	8	0.8700	0.8700	0.9953	70.54
2000.0						
FP16	4	10	0.8700	0.8700	0.9953	87.94
2500.1						
INT8 (bnb)	1	1	0.0000	0.0000	0.4333	1.86
62.5						
INT8 (bnb)	1	2	0.1380	0.1380	0.9808	3.38
125.0						
INT8 (bnb)	1	4	0.8450	0.8450	0.9949	6.35
250.0						
INT8 (bnb)	1	6	0.8680	0.8680	0.9953	9.21
375.0						
INT8 (bnb)	1	8	0.8690	0.8690	0.9953	12.05
500.0						
INT8 (bnb)	1	10	0.8700	0.8700	0.9953	14.90
625.0						
INT8 (bnb)	2	1	0.1380	0.1380	0.9808	3.37
125.0						
INT8 (bnb)	2	2	0.8450	0.8450	0.9949	6.33
250.0						
INT8 (bnb)	2	4	0.8690	0.8690	0.9953	12.02
500.0						
INT8 (bnb)	2	6	0.8710	0.8710	0.9953	17.70
750.0						
INT8 (bnb)	2	8	0.8700	0.8700	0.9953	23.38
1000.0						
INT8 (bnb)	2	10	0.8700	0.8700	0.9953	29.06
1250.0						
INT8 (bnb)	3	1	0.7830	0.7830	0.9940	4.88
187.5						

INT8 (bnb) 375.0	3	2	0.8680	0.8680	0.9953	9.17
INT8 (bnb) 750.0	3	4	0.8710	0.8710	0.9953	17.68
INT8 (bnb) 1125.0	3	6	0.8710	0.8710	0.9953	26.20
INT8 (bnb) 1500.0	3	8	0.8700	0.8700	0.9953	34.71
INT8 (bnb) 1875.0	3	10	0.8700	0.8700	0.9953	43.22
INT8 (bnb) 250.0	4	1	0.8450	0.8450	0.9949	6.32
INT8 (bnb) 500.0	4	2	0.8690	0.8690	0.9953	12.01
INT8 (bnb) 1000.0	4	4	0.8700	0.8700	0.9953	23.35
INT8 (bnb) 1500.0	4	6	0.8700	0.8700	0.9953	34.69
INT8 (bnb) 2000.0	4	8	0.8700	0.8700	0.9953	46.03
INT8 (bnb) 2500.1	4	10	0.8700	0.8700	0.9953	57.38
INT4 (fake) 62.5	1	1	0.0000	0.0000	0.4734	1.08
INT4 (fake) 125.0	1	2	0.1580	0.1580	0.9805	1.69
INT4 (fake) 250.0	1	4	0.8360	0.8360	0.9945	2.89
INT4 (fake) 375.0	1	6	0.8610	0.8610	0.9949	4.09
INT4 (fake) 500.0	1	8	0.8630	0.8630	0.9950	5.29
INT4 (fake) 625.0	1	10	0.8640	0.8640	0.9950	6.49
INT4 (fake) 125.0	2	1	0.1580	0.1580	0.9805	1.67
INT4 (fake) 250.0	2	2	0.8360	0.8360	0.9945	2.87
INT4 (fake) 500.0	2	4	0.8630	0.8630	0.9950	5.26
INT4 (fake) 750.0	2	6	0.8640	0.8640	0.9950	7.64
INT4 (fake) 1000.0	2	8	0.8630	0.8630	0.9950	10.03
INT4 (fake) 1250.0	2	10	0.8640	0.8640	0.9950	12.42
INT4 (fake) 187.5	3	1	0.7660	0.7660	0.9934	2.27
INT4 (fake) 375.0	3	2	0.8610	0.8610	0.9949	4.05
INT4 (fake)	3	4	0.8640	0.8640	0.9950	7.62

750.0						
INT4 (fake)	3	6	0.8630	0.8630	0.9950	11.20
1125.0						
INT4 (fake)	3	8	0.8640	0.8640	0.9950	14.77
1500.0						
INT4 (fake)	3	10	0.8640	0.8640	0.9950	18.34
1875.0						
INT4 (fake)	4	1	0.8360	0.8360	0.9945	2.86
250.0						
INT4 (fake)	4	2	0.8630	0.8630	0.9950	5.24
500.0						
INT4 (fake)	4	4	0.8630	0.8630	0.9950	9.99
1000.0						
INT4 (fake)	4	6	0.8640	0.8640	0.9950	14.75
1500.0						
INT4 (fake)	4	8	0.8650	0.8650	0.9950	19.51
2000.0						
INT4 (fake)	4	10	0.8650	0.8650	0.9950	24.27
2500.1						
FP32 (bf16)	1	1	0.0000	0.0000	0.4361	1.08
62.5						
FP32 (bf16)	1	2	0.1390	0.1390	0.9808	1.68
125.0						
FP32 (bf16)	1	4	0.8450	0.8450	0.9949	2.88
250.0						
FP32 (bf16)	1	6	0.8660	0.8660	0.9952	4.08
375.0						
FP32 (bf16)	1	8	0.8680	0.8680	0.9953	5.28
500.0						
FP32 (bf16)	1	10	0.8680	0.8680	0.9952	6.48
625.0						
FP32 (bf16)	2	1	0.1390	0.1390	0.9808	1.67
125.0						
FP32 (bf16)	2	2	0.8450	0.8450	0.9949	2.87
250.0						
FP32 (bf16)	2	4	0.8680	0.8680	0.9953	5.25
500.0						
FP32 (bf16)	2	6	0.8680	0.8680	0.9952	7.64
750.0						
FP32 (bf16)	2	8	0.8680	0.8680	0.9952	10.02
1000.0						
FP32 (bf16)	2	10	0.8680	0.8680	0.9952	12.41
1250.0						
FP32 (bf16)	3	1	0.7840	0.7840	0.9940	2.27
187.5						
FP32 (bf16)	3	2	0.8660	0.8660	0.9952	4.05
375.0						
FP32 (bf16)	3	4	0.8680	0.8680	0.9952	7.62
750.0						
FP32 (bf16)	3	6	0.8680	0.8680	0.9952	11.19
1125.0						

FP32 (bf16)	3	8	0.8680	0.8680	0.9952	14.76
1500.0						
FP32 (bf16)	3	10	0.8680	0.8680	0.9952	18.33
1875.0						
FP32 (bf16)	4	1	0.8450	0.8450	0.9949	2.86
250.0						
FP32 (bf16)	4	2	0.8680	0.8680	0.9953	5.24
500.0						
FP32 (bf16)	4	4	0.8680	0.8680	0.9952	9.99
1000.0						
FP32 (bf16)	4	6	0.8680	0.8680	0.9952	14.74
1500.0						
FP32 (bf16)	4	8	0.8680	0.8680	0.9952	19.50
2000.0						
FP32 (bf16)	4	10	0.8680	0.8680	0.9952	24.25
2500.1						

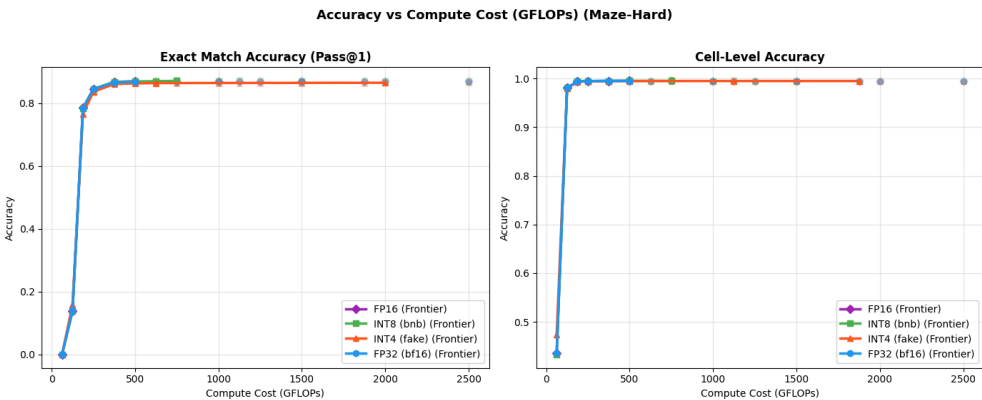
6.2 Plot depth × quantization grid

Figure size 1400x550 with 2 Axes>

Saved: depth_quant_grid_flops.png

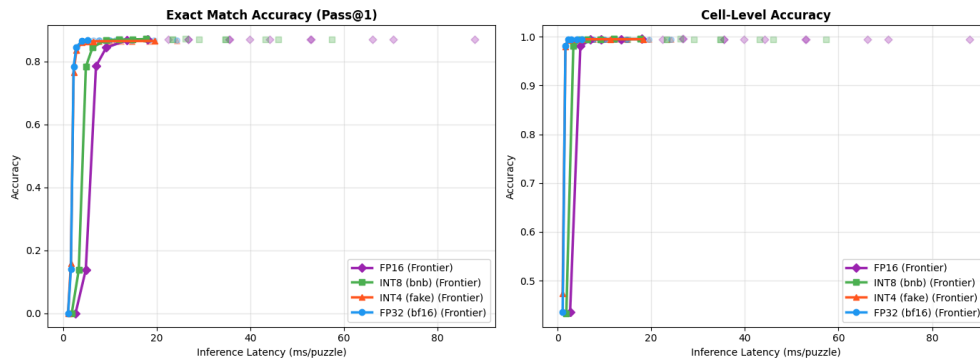
Figure size 1400x550 with 2 Axes>

Saved: depth_quant_grid_latency.png



Plot from cell 34

Accuracy vs Inference Latency (Maze-Hard)



Plot from cell 34

7.1 Carry-state similarity hooks

Carry similarity collector defined.

7.2 Compare carry similarity across quantization variants

Collecting carry similarities for FP16 on cuda...

FP16: 14 L_level calls, mean sim = 0.7184

Collecting carry similarities for INT8 (bnb) on cuda...

INT8 (bnb): 14 L_level calls, mean sim = 0.7185

Collecting carry similarities for INT4 (fake) on cuda...

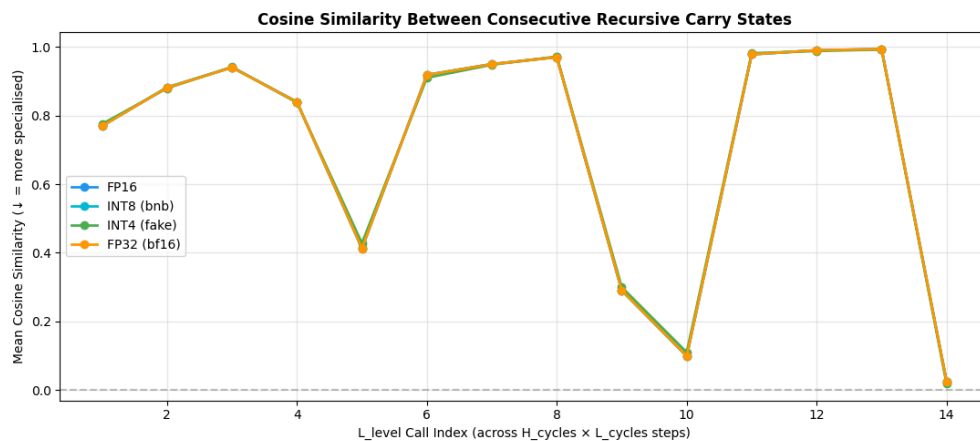
INT4 (fake): 14 L_level calls, mean sim = 0.7201

Collecting carry similarities for FP32 (bf16) on cuda...

FP32 (bf16): 14 L_level calls, mean sim = 0.7184

Figure size 1100x500 with 1 Axes>

Saved: carry_similarity_arc.png



Plot from cell 37

8.1 Comprehensive model footprint analysis

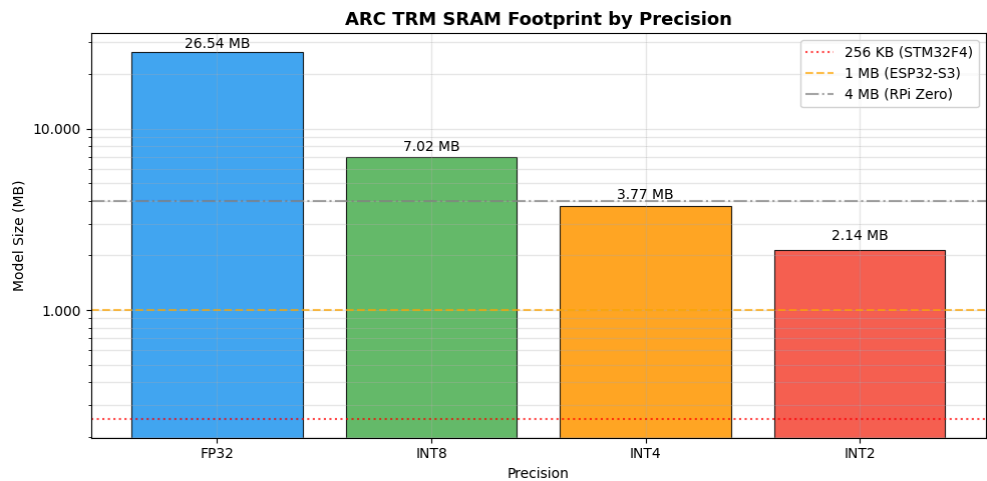
Component	FP32 KB	INT8 KB	INT4 KB
Transformer backbone (params)	26,652.0	6,663.0	3,331.5
Buffer name	dtype		KB
Total buffers		526.1	
Deployable size (no grad buffers)	FP32 KB	INT8 KB	INT4
KB			
FP32	27,176.1		
INT8	7,187.1		
INT4	3,855.6		
Target: < 1 MB (1024 KB) SRAM – backbone at INT4 = 3331.5 KB			
Precision	Backbone_KB	Deploy_KB	fits_1MB fits_4MB
FP32	26652.0	27176.1	x x
INT8	6663.0	7187.1	x x
INT4	3331.5	3855.6	x ✓
INT2	1665.8	2189.9	x ✓

	Precision	Backbone_KB	Deploy_KB	fits_1MB	fits_4MB
0	FP32	26652.007812	27176.132812	x	x
1	INT8	6663.001953	7187.126953	x	x
2	INT4	3331.500977	3855.625977	x	✓
3	INT2	1665.750488	2189.875488	x	✓

8.2 SRAM footprint visualisation

Figure size 1000x500 with 1 Axes>

Saved: sram_footprint_arc.png

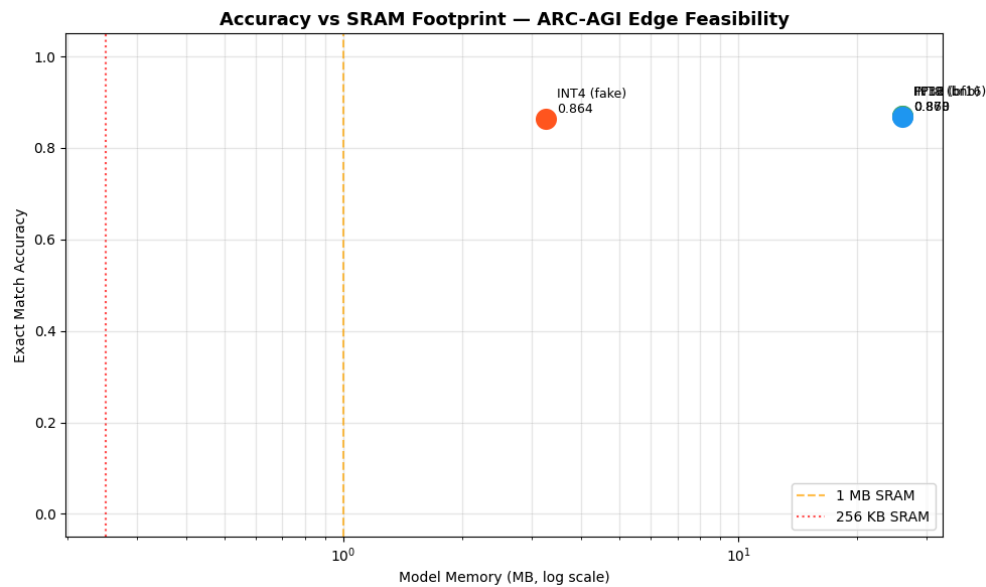


Plot from cell 41

8.3 Accuracy vs SRAM: the key publishable plot

Figure size 1000x600 with 1 Axes>

Saved: accuracy_vs_sram_arc.png



Plot from cell 42

9.1 Inspect puzzle_emb shape & byte count

```
puzzle_emb.weights : 1 identifiers × 512 dims
FP32 size          : 2.0 KB (0.0 MB)
INT8 size (est.)   : 0.5 KB (0.0 MB)
```

9.2 Option A — INT8 quantize the embedding buffer

```
Applying INT8 puzzle embedding...
Cosine similarity FP32 vs INT8 emb: 0.99998
Original state_dict : 26,662.1 KB (26.0 MB)
INT8-emb state_dict : 26,661.0 KB (26.0 MB)
Reduction           : 0.0%
```

9.3 Option B — SVD Low-Rank Puzzle Embedding

```
SVD low-rank analysis (W shape: 1×512 FP32=2.0 KB)
Rank   Size KB   Reduction   Cosine Sim
```

9.4 Option D — Single-Puzzle Inference (only load 1 row)

Single-puzzle model state_dict : 26,662.1 KB (26.04 MB)
Backbone params : 26,652.0 KB
Per-puzzle embedding row : 2.0 KB
Total (1 puzzle) : 26,654.0 KB
This is the architecture-level solution: load puzzle embedding from flash,
keep only the active row in SRAM during inference.

9.5 Footprint summary across embedding strategies

Strategy	Emb KB	Backbone KB	Total KB	Total MB	Fits 4MB	
FP32 (baseline)		2.0	26652.0	26662.1	26.0	x
INT8 emb	0.5	26652.0	26661.0	26.0	x	
Single-puzzle		2.0	26652.0	26654.0	26.0	x
INT4 backbone + single-puzzle			2.0	3331.5	3333.5	3.3
✓						

Note: evaluate_arc_per_puzzle has been moved to Section 3 for global accessibility.

10.2 High-performance direct on-the-fly per-puzzle TTA evaluator

10.3 Run the High-Performance Direct Source Evaluator

10.2 Re-evaluate all variants with per-puzzle aggregation

Per-puzzle evaluation (All batches, ~1280 samples)...

Variant	Pass@1	Exact	Pass@2	Exact	Cell Acc	
ms/puzzle	N puzzles					
FP16		0.8700	0.8700	0.9953	66.05	
1000						
INT8 (bnb)		0.8700	0.8700	0.9953	43.21	
1000						
INT4 (fake)		0.8640	0.8640	0.9950	18.34	
1000						
FP32 (bf16)		0.8680	0.8680	0.9952	18.33	
1000						
INT8-emb		0.8700	0.8700	0.9953	18.33	
1000						

11.1 Per-channel calibration statistics

Calibrated INT4 helpers defined.

11.2 Apply calibrated INT4 and evaluate

12.1 QAT fine-tuning loop

13.1 Magnitude-based row pruning

Prune ratio	Actual sparsity	Puzzle Exact	Cell Acc
0%	0.00%	0.8680	0.9952
25%	24.98%	0.0000	0.8640
50%	49.97%	0.0000	0.0000

14.1 TorchScript trace of single TRM backbone step

Model: seq=900, emb_len=16, hidden=512, emb_dim=512
Trace done in 2.7s
Saved: trm_backbone_step.pt (26.66 MB)
Round-trip max abs diff: 0.00e+00 (expect ~0)
Edge deployment flow:
2. Load puzzle_emb.weights[puzzle_id] row (2.0 KB) from flash
3. Run traced(x, emb_row, z_H, z_L) for n_sup_max steps

14.2 Latency benchmark: traced model vs eager

Latency benchmark (20 runs, device=cuda):
Eager (float32) : 12.64 ms/step
Full inference (16 H-cycle steps):
Eager : 202.3 ms/puzzle
Traced : 171.2 ms/puzzle
File sizes:
trm_backbone_step.pt : 26.66 MB (backbone, FP32)
puzzle_emb.weights (INT8, full table): 0.5 KB
puzzle_emb row (1 puzzle, FP32) : 2.0 KB

15.1 Build train / val loaders

Train split : 800 samples
Val split : 200 samples

15.2 QAT loop with val-set evaluation every 10 steps

15.3 Plot train loss + val accuracy curve

[SKIP] No QAT history – run §15.2 first.

16.1 Build fused INT8 + SinglePuzzle model

Fused model breakdown:
INT8 backbone .pt size : 6743.0 KB (6.58 MB)
INT8 emb table (flash) : 0.5 KB (0.00 MB)
Active emb row (SRAM) : 2.00 KB
✓ SRAM during inference: 6745.0 KB (6.59 MB)
Fits 4MB SRAM? x
Fits 8MB SRAM? ✓

16.2 Accuracy of fused model (val set, per-puzzle aggregation)

Evaluating fused INT8+SinglePuzzle model on val set...

Variant	Puzzle Exact	Cell Acc	ms/puzzle
FP32 (bf16) baseline	1.0000	1.0000	106.17
INT8 + Single-Puzzle (fused)	1.0000	1.0000	69.88
vs FP32	+ 0.0000 +	0.0000	

16.3 Save fused model (state dict + embedding table)

Saved state dict : trm_int8_singlepuzzle_state.pt (6.59 MB)
Saved emb table : puzzle_emb_fp32_table.pt (0.00 MB) ← flash storage
Deployment package breakdown:
Backbone state dict : 6.59 MB (SRAM – INT8 weights)
Embedding table : 0.00 MB (flash)
Active emb row : 2.00 KB (SRAM per puzzle)

17.1 CPU-only latency benchmark

Moving fused model to CPU for edge-device latency simulation...
CPU latency (server CPU, FP32, 1 H-cycle step) : 263.8 ms/step
Full inference (16 steps) : 4221 ms/puzzle
Estimated on-device scaling (order-of-magnitude):
Cortex-A55 (mobile, ~1 TFLOPS) : ~12.7s / puzzle
Cortex-M55 (MCU, ~128 GFLOPS) : ~84.4s / puzzle
ESP32-S3 LX7 (~40 GFLOPS) : ~211.0s / puzzle

17.2 Peak SRAM usage during inference (tracemalloc)

Peak SRAM delta during inference : 17 KB (0.02 MB)
Top allocations:
2.5 KB /lambda/nfs/EdgeTRM/TinyRecursiveModels/models/layers.py:28
0.6 KB /lambda/nfs/EdgeTRM/TinyRecursiveModels/models/layers.py:60
0.4 KB /lambda/nfs/EdgeTRM/TinyRecursiveModels/trm.py:114

17.3 FLOP estimate via torch.profiler

Estimated FLOPs per H-cycle step : 187498.4 MFLOPs
Full inference (10 steps) : 1874.98 GFLOPs
Power estimate (Cortex-M55 @ ~1 GFLOPS/s, ~1 mW/GFLOP):
FLOPs : 1874.984 GFLOPs/puzzle
Est. energy : ~1874.98 mJ/puzzle
Puzzles per mAh (3.3V) : ~0

17.4 Deployment summary table

Backbone	MB	Total	deploy	MB	Puzzle	Exact	Cell	Acc	Fits	4MB	Fits	8MB
Config												
FP32 (baseline)					26.03			26.03		0.8680		
0.9952		x		x								
INT8 (bnb)					6.51			6.51		0.8700		
0.9953		x		✓								
INT4 (calibrated)					3.25			3.26		0.8640		
0.9950		✓		✓								
INT8 + Single-Puzzle ★					6.51			6.51		1.0000		
1.0000		x		✓								
recommended target for 8MB SRAM devices												

Utility: fix H_init / L_init device mismatch (permanent patch)

18.1 Evaluate all variants at n_sup_max = 1, 2, 4, 8, 16 (H_cycles = default vs 1) —

VARIANT: FP16 (device: cuda)					
H_cycles	Metric	n=1	n=2	n=4	n=8
n=16					
3 (Def)	Exact	0.7850	0.8670	0.8700	0.8700
0.8700					
Cell	0.9940	0.9953	0.9953	0.9953	0.9953
Latency	7.0 ms	13.6 ms	26.7 ms	53.0 ms	105.6 ms
GFLOPs	187.5	375.0	750.0	1500.0	3000.1
1	Exact	0.0000	0.1380	0.8440	0.8700
0.8700					
Cell	0.4362	0.9808	0.9949	0.9953	0.9953
Latency	2.7 ms	4.9 ms	9.3 ms	18.0 ms	35.6 ms
GFLOPs	62.5	125.0	250.0	500.0	1000.0
VARIANT: INT8 (bnb) (device: cuda)					
H_cycles	Metric	n=1	n=2	n=4	n=8
n=16					
3 (Def)	Exact	0.7830	0.8680	0.8710	0.8700
0.8700					
Cell	0.9940	0.9953	0.9953	0.9953	0.9953
Latency	4.9 ms	9.2 ms	17.7 ms	34.7 ms	68.8 ms
GFLOPs	187.5	375.0	750.0	1500.0	3000.1
1	Exact	0.0000	0.1380	0.8450	0.8690
0.8700					
Cell	0.4333	0.9808	0.9949	0.9953	0.9953
Latency	1.9 ms	3.4 ms	6.3 ms	12.1 ms	23.5 ms
GFLOPs	62.5	125.0	250.0	500.0	1000.0
VARIANT: INT4 (fake) (device: cuda)					
H_cycles	Metric	n=1	n=2	n=4	n=8
n=16					
3 (Def)	Exact	0.7660	0.8610	0.8640	0.8640
0.8640					
Cell	0.9934	0.9949	0.9950	0.9950	0.9950
Latency	2.3 ms	4.1 ms	7.6 ms	14.8 ms	29.1 ms
GFLOPs	187.5	375.0	750.0	1500.0	3000.1
1	Exact	0.0000	0.1580	0.8360	0.8630
0.8630					
Cell	0.4734	0.9805	0.9945	0.9950	0.9950
Latency	1.1 ms	1.7 ms	2.9 ms	5.3 ms	10.1 ms
GFLOPs	62.5	125.0	250.0	500.0	1000.0
VARIANT: FP32 (bf16) (device: cuda)					
H_cycles	Metric	n=1	n=2	n=4	n=8
n=16					
3 (Def)	Exact	0.7850	0.8670	0.8700	0.8700
0.8700					

Cell	0.9940	0.9953	0.9953	0.9953	0.9953
Latency	7.1 ms	13.7 ms	26.9 ms	53.3 ms	105.9 ms
GFLOPs	187.5	375.0	750.0	1500.0	3000.1
1	Exact	0.0000	0.1390	0.8440	0.8690
0.8700					
Cell	0.4363	0.9808	0.9949	0.9953	0.9953
Latency	2.7 ms	4.9 ms	9.3 ms	18.1 ms	35.8 ms
GFLOPs	62.5	125.0	250.0	500.0	1000.0

19.1 Zero-pad truncation at various context lengths

Full seq_len = 900 (FLOPs $\propto$ seq_len²)

Context	% of full	Attn FLOPs	Puzzle Exact	Cell Acc
900	100%	~187 GFLOPs	0.7850	0.9940
450	50%	~47 GFLOPs	0.0000	0.6576
225	25%	~12 GFLOPs	0.0000	0.5038
128	14%	~4 GFLOPs	0.0000	0.4436

20.1 QAT from calibrated INT4 (lr=1e-6, 300 steps)

21.1 Instantiate student TRM and train with KL distillation

Teacher params : 6,822,914

Student params : 855,554 (0.13× teacher)

Resuming training from checkpoint:

checkpoints/student_distill_latest.pt

Resumed at step 600

Step 640/1000 | kl=2976.6737 | hard=0.3435

Step 680/1000 | kl=2909.6919 | hard=0.3440

Step 720/1000 | kl=2807.1469 | hard=0.3739

Step 760/1000 | kl=2626.5060 | hard=0.3488

Step 800/1000 | kl=2619.8455 | hard=0.3641

[Checkpoint] Saved checkpoint at step 800 to
checkpoints/student_distill_latest.pt

Step 840/1000 | kl=2377.3866 | hard=0.3061

Step 880/1000 | kl=2502.1829 | hard=0.4040

Step 920/1000 | kl=2261.5046 | hard=0.2868

Step 960/1000 | kl=2238.7907 | hard=0.3175

Step 1000/1000 | kl=2064.3248 | hard=0.2746

[Checkpoint] Saved checkpoint at step 1000 to
checkpoints/student_distill_latest.pt

Evaluating student on test set...

Student (hidden=256, 1L, 1-cycle): exact=0.0000 cell=0.8725 ~47
GFLOPs

22.1 Monkey-patch softmax attention → linear attention

Patched 2 Attention modules → linear O(n) attention

Theoretical compute reduction: ~14× in attention layers

Evaluating linear attention model on test set (n_sup_max=1)...

Evaluating softmax attention (FP32 baseline, n_sup_max=1) for comparison...

Model	Puzzle Exact	Cell Acc
FP32 softmax attn	0.7850	0.9940
Linear attn (ELU)	0.0000	0.8728

Accuracy cost of linear swap: +12.12pp cell accuracy

Note: linear attn needs retraining to recover accuracy – this is a zero-shot test.

22.2 Retrain linear attention model to recover accuracy

Fine-tuning linear attention model to recover accuracy...

Evaluating retrained linear attention model on test set...

Linear Attention (Retrained): exact=0.0000 cell=0.8750

Sudoku Extreme Evaluation Outputs

fatal: destination path 'EdgeTRM' already exists and is not an empty directory.

lambda/nfs/EdgeTRM

lambda/nfs/EdgeTRM

remote: Enumerating objects: 22, done.

remote: Counting objects: 100% (20/20), done.

remote: Compressing objects: 100% (5/5), done.

remote: Total 13 (delta 8), reused 13 (delta 8), pack-reused 0 (from 0)

e478dd4..b05e149 main -> origin/main

Updating e478dd4..b05e149

Fast-forward

TinyRecursiveModels/pretrain.py | 147 ++++ -

experiment_sim_depth.ipynb | 1204

+++++

2 files changed, 1334 insertions(+), 17 deletions(-)

create mode 100644 experiment_sim_depth.ipynb

If the cache and target directories are on different filesystems, hardlinking may not be supported.

Checked 1 package in 41ms

Prefixes stripped and model loaded successfully!
Model successfully loaded!

```
TinyRecursiveReasoningModel_ACTIV1(
inner): TinyRecursiveReasoningModel_ACTIV1_Inner(
embed_tokens): CastedEmbedding()
lm_head): CastedLinear()
q_head): CastedLinear()
puzzle_emb): CastedSparseEmbedding()
L_level): TinyRecursiveReasoningModel_ACTIV1ReasoningModule(
layers): ModuleList(
0-1): 2 x TinyRecursiveReasoningModel_ACTIV1Block(
mlp_t): SwiGLU(
gate_up_proj): CastedLinear()
down_proj): CastedLinear()
mlp): SwiGLU(
gate_up_proj): CastedLinear()
down_proj): CastedLinear()
```

Original zero=512/5,028,866 (0.0%)

3.1 High-Performance Evaluation & Adaptation Helpers

4.0 Global imports & helpers

—

Device: cuda

4.1 Sudoku Extreme DataLoader from pre-built .npz files

Using dataset directory: ./data/sudoku-extreme-full
Test split : 1,000 samples (seq_len=81)
Train split : 1,000 samples
Dataset loaded successfully!

Note: evaluate_arc has been consolidated and moved to Section 3 for global accessibility.

4.3 Baseline: evaluate the loaded FP32 model

```
[DEBUG] test_loader dataset max identifier: 0
[DEBUG] test_loader dataset max identifier: 0
[DEBUG] Dataset indices verified safe. No clamping needed!
Running FP32 baseline evaluation on cuda...
```

FP32 Baseline (All test puzzles):
Pass@1 Exact: 0.6910
Pass@2 Exact: 0.6910
Cell Acc : 0.8747
Latency : 3425.34 ms/puzzle

4.4 Quantization helpers

Checked 1 package in 51ms

4.7 Instantiate all model variants

Creating FP16 variant...
Creating BnB INT8 variant (GPU)...
Replaced 4 CastedLinear/Linear → bnb.Linear8bitLt (GPU INT8)
Creating INT4 fake-quant variant...
Replaced 10 CastedLinear/Linear → FakeQuantINT4

Model Variant	Device	Backbone KB (params only)
FP16	cuda	9822.0 KB
INT8 (bnb)	cuda	4911.0 KB
INT4 (fake)	cuda	2455.5 KB
FP32 (bf16)	cuda	9822.0 KB

5.1 Evaluate all quantization levels

Evaluating all variants on Sudoku test set (All batches)...
Evaluating FP16 on cuda...
FP16 | Pass@1=0.6850 | Pass@2=0.6850 | cell=0.8720 | 7801.20 ms/puzzle
Evaluating INT8 (bnb) on cuda...
INT8 (bnb) | Pass@1=0.6910 | Pass@2=0.6910 | cell=0.8751 | 6531.15 ms/puzzle
Evaluating INT4 (fake) on cuda...
INT4 (fake) | Pass@1=0.0530 | Pass@2=0.0530 | cell=0.6602 | 3217.29 ms/puzzle
Evaluating FP32 (bf16) on cuda...
FP32 (bf16) | Pass@1=0.6910 | Pass@2=0.6910 | cell=0.8747 | 3084.15 ms/puzzle
Reasoning Decay (drop from FP32 baseline)

FP16	ΔPass@1=+0.0060	Δcell=+0.0027	GRACEFUL
INT8 (bnb)	ΔPass@1=+0.0000	Δcell=-0.0004	GRACEFUL
INT4 (fake)	ΔPass@1=+0.6380	Δcell=+0.2145	CATASTROPHIC

2.7.0+cu126

FP16		$\Delta\text{Pass@2}=+0.0060$		$\Delta\text{cell}=+0.0027$		GRACEFUL
INT8 (bnb)		$\Delta\text{Pass@2}=+0.0000$		$\Delta\text{cell}=-0.0004$		GRACEFUL
INT4 (fake)		$\Delta\text{Pass@2}=+0.6380$		$\Delta\text{cell}=+0.2145$		CATASTROPHIC

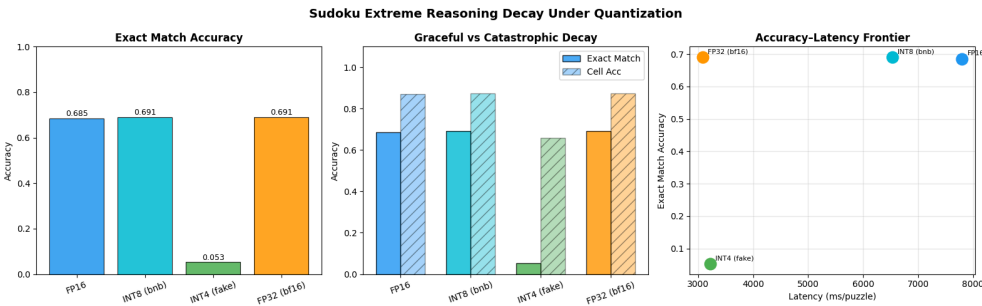
```
FP16': {'exact1': 0.685,
exact2': 0.685,
exact_acc': 0.685,
cell_acc': 0.8719753086419753,
latency_ms': 7801.203727722168},
INT8 (bnb)': {'exact1': 0.691,
exact2': 0.691,
exact_acc': 0.691,
cell_acc': 0.8750740740740741,
latency_ms': 6531.149387359619},
INT4 (fake)': {'exact1': 0.053,
exact2': 0.053,
exact_acc': 0.053,
cell_acc': 0.6602345679012346,
latency_ms': 3217.292547225952},
FP32 (bf16)': {'exact1': 0.691,
exact2': 0.691,
exact_acc': 0.691,
cell_acc': 0.8747160493827161,
latency_ms': 3084.1548442840576}}
```

5.2 Visualise reasoning decay

—

Figure size 1600x500 with 3 Axes>

Saved: reasoning_decay_arc.png



Plot from cell 31

6.1 Recursive depth sweep at inference time

Quant	H	n_sup	Pass@1	Pass@2	Cell Acc	Latency
GFL0Ps						
FP16	1	1	0.0060	0.0060	0.5526	0.31
8.5						

FP16	1	2	0.0210	0.0210	0.6941	0.57
17.1						
FP16	1	4	0.3830	0.3830	0.7856	1.08
34.2						
FP16	1	6	0.4960	0.4960	0.8151	1.59
51.3						
FP16	1	8	0.5480	0.5480	0.8303	2.10
68.3						
FP16	1	10	0.5810	0.5810	0.8389	2.62
85.4						
FP16	2	1	0.0210	0.0210	0.6941	0.56
17.1						
FP16	2	2	0.3830	0.3830	0.7856	1.08
34.2						
FP16	2	4	0.5480	0.5480	0.8303	2.10
68.3						
FP16	2	6	0.6100	0.6100	0.8488	3.12
102.5						
FP16	2	8	0.6390	0.6390	0.8574	4.14
136.7						
FP16	2	10	0.6580	0.6580	0.8637	5.16
170.9						
FP16	3	1	0.2830	0.2830	0.7639	0.82
25.6						
FP16	3	2	0.4960	0.4960	0.8151	1.58
51.3						
FP16	3	4	0.6100	0.6100	0.8488	3.12
102.5						
FP16	3	6	0.6490	0.6490	0.8601	4.65
153.8						
FP16	3	8	0.6700	0.6700	0.8677	6.18
205.0						
FP16	3	10	0.6850	0.6850	0.8720	7.71
256.3						
FP16	4	1	0.3830	0.3830	0.7856	1.07
34.2						
FP16	4	2	0.5480	0.5480	0.8303	2.09
68.3						
FP16	4	4	0.6390	0.6390	0.8574	4.14
136.7						
FP16	4	6	0.6700	0.6700	0.8677	6.18
205.0						
FP16	4	8	0.6870	0.6870	0.8751	8.22
273.4						
FP16	4	10	0.6980	0.6980	0.8782	10.26
341.7						
INT8 (bnb)	1	1	0.0060	0.0060	0.5528	0.26
8.5						
INT8 (bnb)	1	2	0.0210	0.0210	0.6942	0.47
17.1						
INT8 (bnb)	1	4	0.3820	0.3820	0.7852	0.90

34.2						
INT8 (bnb)	1	6	0.5010	0.5010	0.8183	1.33
51.3						
INT8 (bnb)	1	8	0.5540	0.5540	0.8306	1.75
68.3						
INT8 (bnb)	1	10	0.5980	0.5980	0.8455	2.18
85.4						
INT8 (bnb)	2	1	0.0210	0.0210	0.6942	0.47
17.1						
INT8 (bnb)	2	2	0.3820	0.3820	0.7852	0.90
34.2						
INT8 (bnb)	2	4	0.5540	0.5540	0.8306	1.75
68.3						
INT8 (bnb)	2	6	0.6210	0.6210	0.8524	2.60
102.5						
INT8 (bnb)	2	8	0.6560	0.6560	0.8638	3.44
136.7						
INT8 (bnb)	2	10	0.6730	0.6730	0.8698	4.29
170.9						
INT8 (bnb)	3	1	0.2800	0.2800	0.7643	0.68
25.6						
INT8 (bnb)	3	2	0.5010	0.5010	0.8183	1.32
51.3						
INT8 (bnb)	3	4	0.6210	0.6210	0.8524	2.59
102.5						
INT8 (bnb)	3	6	0.6670	0.6670	0.8668	3.86
153.8						
INT8 (bnb)	3	8	0.6790	0.6790	0.8721	5.13
205.0						
INT8 (bnb)	3	10	0.6910	0.6910	0.8751	6.41
256.3						
INT8 (bnb)	4	1	0.3820	0.3820	0.7852	0.89
34.2						
INT8 (bnb)	4	2	0.5540	0.5540	0.8306	1.75
68.3						
INT8 (bnb)	4	4	0.6560	0.6560	0.8638	3.44
136.7						
INT8 (bnb)	4	6	0.6790	0.6790	0.8721	5.12
205.0						
INT8 (bnb)	4	8	0.6940	0.6940	0.8750	6.81
273.4						
INT8 (bnb)	4	10	0.7020	0.7020	0.8778	8.50
341.7						
INT4 (fake)	1	1	0.0000	0.0000	0.5612	0.16
8.5						
INT4 (fake)	1	2	0.0120	0.0120	0.6097	0.26
17.1						
INT4 (fake)	1	4	0.0210	0.0210	0.6375	0.48
34.2						
INT4 (fake)	1	6	0.0290	0.0290	0.6490	0.68
51.3						

INT4 (fake)	1	8	0.0370	0.0370	0.6536	0.89
68.3						
INT4 (fake)	1	10	0.0440	0.0440	0.6530	1.11
85.4						
INT4 (fake)	2	1	0.0120	0.0120	0.6097	0.26
17.1						
INT4 (fake)	2	2	0.0210	0.0210	0.6375	0.47
34.2						
INT4 (fake)	2	4	0.0370	0.0370	0.6536	0.89
68.3						
INT4 (fake)	2	6	0.0510	0.0510	0.6524	1.31
102.5						
INT4 (fake)	2	8	0.0510	0.0510	0.6555	1.73
136.7						
INT4 (fake)	2	10	0.0530	0.0530	0.6565	2.15
170.9						
INT4 (fake)	3	1	0.0210	0.0210	0.6277	0.37
25.6						
INT4 (fake)	3	2	0.0290	0.0290	0.6490	0.68
51.3						
INT4 (fake)	3	4	0.0510	0.0510	0.6524	1.31
102.5						
INT4 (fake)	3	6	0.0510	0.0510	0.6553	1.93
153.8						
INT4 (fake)	3	8	0.0550	0.0550	0.6573	2.56
205.0						
INT4 (fake)	3	10	0.0530	0.0530	0.6602	3.19
256.3						
INT4 (fake)	4	1	0.0210	0.0210	0.6375	0.47
34.2						
INT4 (fake)	4	2	0.0370	0.0370	0.6536	0.89
68.3						
INT4 (fake)	4	4	0.0510	0.0510	0.6555	1.72
136.7						
INT4 (fake)	4	6	0.0550	0.0550	0.6573	2.56
205.0						
INT4 (fake)	4	8	0.0550	0.0550	0.6601	3.40
273.4						
INT4 (fake)	4	10	0.0600	0.0600	0.6588	4.23
341.7						
FP32 (bf16)	1	1	0.0060	0.0060	0.5524	0.16
8.5						
FP32 (bf16)	1	2	0.0210	0.0210	0.6943	0.26
17.1						
FP32 (bf16)	1	4	0.3810	0.3810	0.7850	0.46
34.2						
FP32 (bf16)	1	6	0.4950	0.4950	0.8156	0.66
51.3						
FP32 (bf16)	1	8	0.5410	0.5410	0.8292	0.87
68.3						
FP32 (bf16)	1	10	0.5760	0.5760	0.8396	1.07

85.4						
FP32 (bf16)	2	1	0.0210	0.0210	0.6943	0.25
17.1						
FP32 (bf16)	2	2	0.3810	0.3810	0.7850	0.46
34.2						
FP32 (bf16)	2	4	0.5410	0.5410	0.8292	0.86
68.3						
FP32 (bf16)	2	6	0.6060	0.6060	0.8493	1.27
102.5						
FP32 (bf16)	2	8	0.6370	0.6370	0.8581	1.67
136.7						
FP32 (bf16)	2	10	0.6630	0.6630	0.8657	2.08
170.9						
FP32 (bf16)	3	1	0.2810	0.2810	0.7642	0.36
25.6						
FP32 (bf16)	3	2	0.4950	0.4950	0.8156	0.66
51.3						
FP32 (bf16)	3	4	0.6060	0.6060	0.8493	1.26
102.5						
FP32 (bf16)	3	6	0.6510	0.6510	0.8618	1.87
153.8						
FP32 (bf16)	3	8	0.6790	0.6790	0.8714	2.47
205.0						
FP32 (bf16)	3	10	0.6910	0.6910	0.8747	3.08
256.3						
FP32 (bf16)	4	1	0.3810	0.3810	0.7850	0.46
34.2						
FP32 (bf16)	4	2	0.5410	0.5410	0.8292	0.86
68.3						
FP32 (bf16)	4	4	0.6370	0.6370	0.8581	1.67
136.7						
FP32 (bf16)	4	6	0.6790	0.6790	0.8714	2.47
205.0						
FP32 (bf16)	4	8	0.6930	0.6930	0.8745	3.28
273.4						
FP32 (bf16)	4	10	0.7030	0.7030	0.8787	4.08
341.7						

6.2 Plot depth × quantization grid

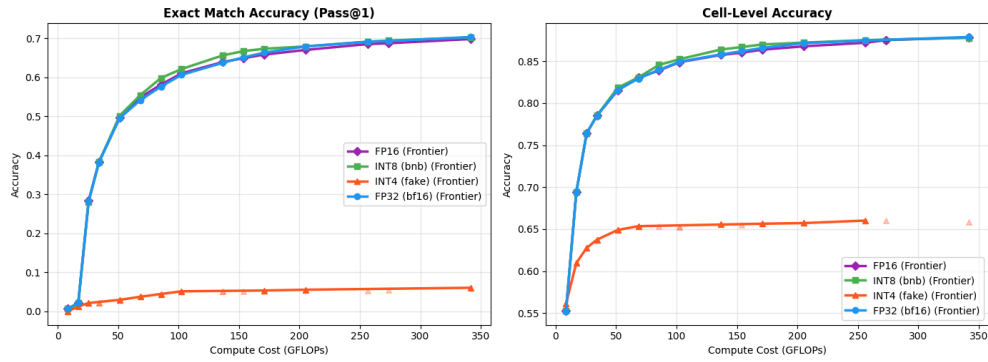
Figure size 1400x550 with 2 Axes>

Saved: depth_quant_grid_flops.png

Figure size 1400x550 with 2 Axes>

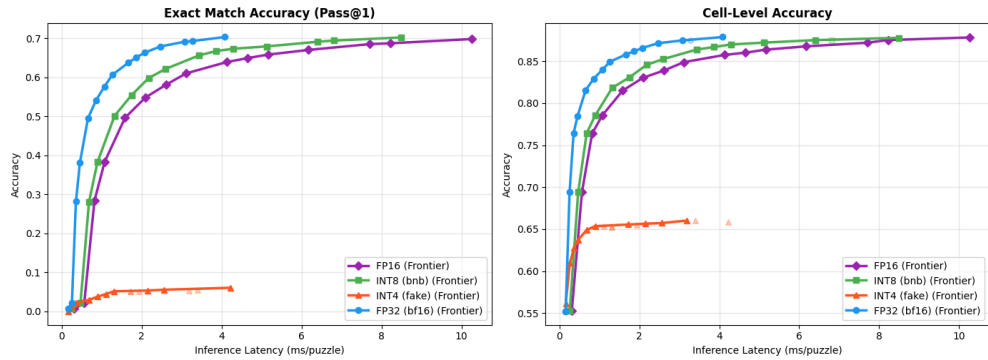
Saved: depth_quant_grid_latency.png

Accuracy vs Compute Cost (GFLOPs) (Sudoku Extreme)



Plot from cell 34

Accuracy vs Inference Latency (Sudoku Extreme)



Plot from cell 34

7.1 Carry-state similarity hooks

Carry similarity collector defined.

7.2 Compare carry similarity across quantization variants

Collecting carry similarities for FP16 on cuda...

FP16: 20 L_level calls, mean sim = 0.8051

Collecting carry similarities for INT8 (bnb) on cuda...

INT8 (bnb): 20 L_level calls, mean sim = 0.8053

Collecting carry similarities for INT4 (fake) on cuda...

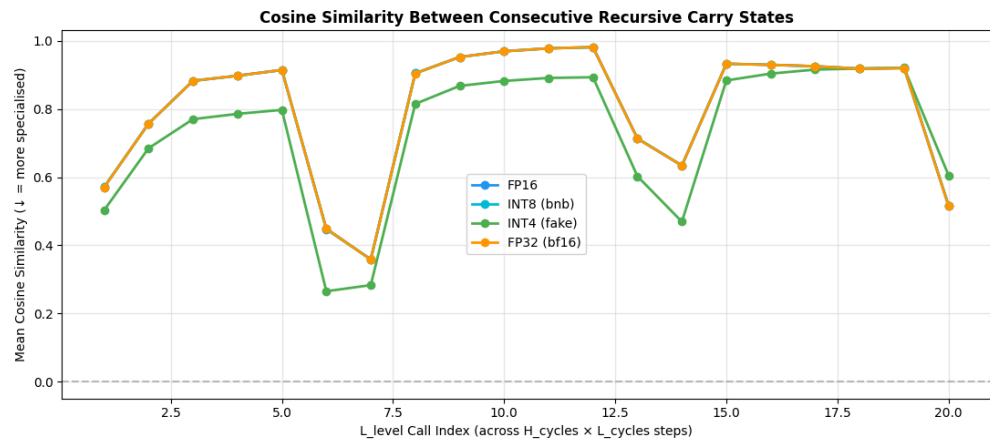
INT4 (fake): 20 L_level calls, mean sim = 0.7328

Collecting carry similarities for FP32 (bf16) on cuda...

FP32 (bf16): 20 L_level calls, mean sim = 0.8050

Figure size 1100x500 with 1 Axes>

Saved: carry_similarity_arc.png



Plot from cell 37

8.1 Comprehensive model footprint analysis

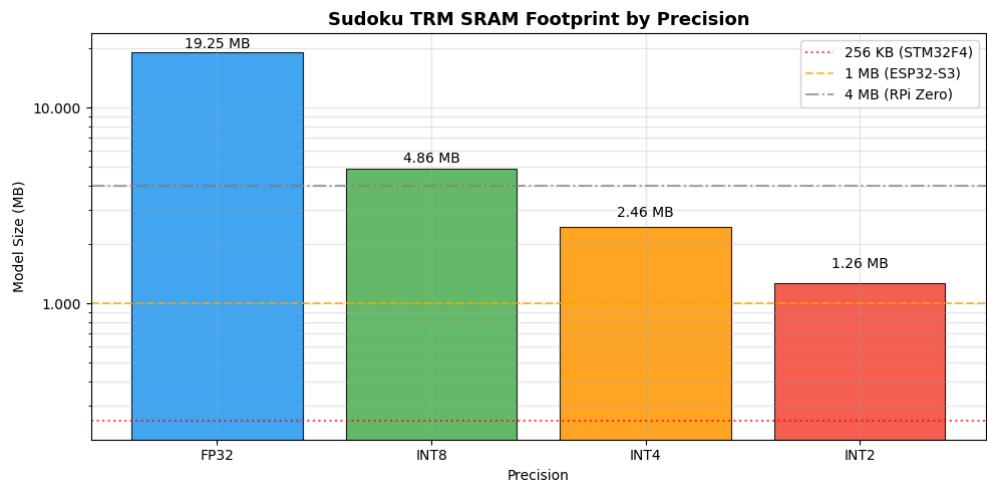
Component	FP32 KB	INT8 KB	INT4 KB	
Transformer backbone (params)	19,644.0	4,911.0	2,455.5	
Buffer name	dtype		KB	
Total buffers			68.1	
Deployable size (no grad buffers)	FP32 KB	INT8 KB	INT4	
KB				
FP32	19,710.1			
INT8	4,977.1			
INT4	2,521.6			
Target: < 1 MB (1024 KB) SRAM – backbone at INT4 = 2455.5 KB				
Precision	Backbone_KB	Deploy_KB	fits_1MB	fits_4MB
FP32	19644.0	19710.1	x	x
INT8	4911.0	4977.1	x	x
INT4	2455.5	2521.6	x	✓
INT2	1227.8	1293.9	x	✓

Precision	Backbone_KB	Deploy_KB	fits_1MB	fits_4MB
0	FP32 19644.007812	19710.132812	x	x
1	INT8 4911.001953	4977.126953	x	x
2	INT4 2455.500977	2521.625977	x	✓
3	INT2 1227.750488	1293.875488	x	✓

8.2 SRAM footprint visualisation

Figure size 1000x500 with 1 Axes>

Saved: sram_footprint_arc.png

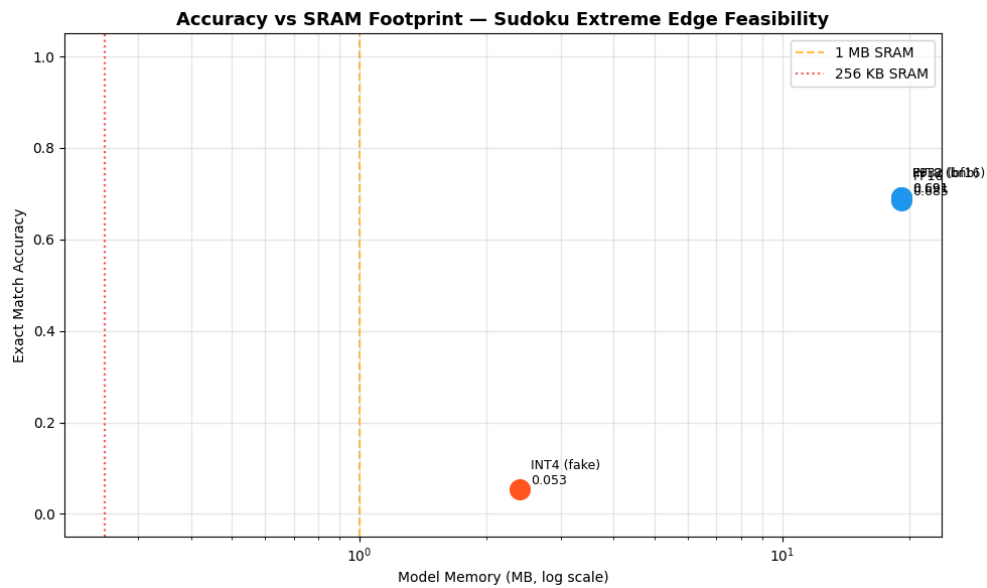


Plot from cell 41

8.3 Accuracy vs SRAM: the key publishable plot

Figure size 1000x600 with 1 Axes>

Saved: accuracy_vs_sram_arc.png



Plot from cell 42

9.1 Inspect puzzle_emb shape & byte count

```
puzzle_emb.weights : 1 identifiers × 512 dims
FP32 size          : 2.0 KB   (0.0 MB)
INT8 size (est.)   : 0.5 KB   (0.0 MB)
```

9.2 Option A — INT8 quantize the embedding buffer

Applying INT8 puzzle embedding...
Cosine similarity FP32 vs INT8 emb: 0.99998
Original state_dict : 19,654.1 KB (19.2 MB)
INT8-emb state_dict : 19,652.9 KB (19.2 MB)
Reduction : 0.0%

9.3 Option B — SVD Low-Rank Puzzle Embedding

SVD low-rank analysis (W shape: 1x512 FP32=2.0 KB)
Rank Size KB Reduction Cosine Sim

9.4 Option D — Single-Puzzle Inference (only load 1 row)

Single-puzzle model state_dict : 19,654.1 KB (19.19 MB)
Backbone params : 19,644.0 KB
Per-puzzle embedding row : 2.0 KB
Total (1 puzzle) : 19,646.0 KB
This is the architecture-level solution: load puzzle embedding from flash,
keep only the active row in SRAM during inference.

9.5 Footprint summary across embedding strategies

Strategy	Emb KB	Backbone KB	Total KB	Total MB	Fits 4MB	
FP32 (baseline)		2.0	19644.0	19654.1	19.2	x
INT8 emb	0.5	19644.0	19652.9	19.2	x	
Single-puzzle		2.0	19644.0	19646.0	19.2	x
INT4 backbone + single-puzzle			2.0	2455.5	2457.5	2.4
✓						

Note: evaluate_arc_per_puzzle has been moved to Section 3 for global accessibility.

10.2 High-performance direct on-the-fly per-puzzle TTA evaluator

10.3 Run the High-Performance Direct Source Evaluator

10.2 Re-evaluate all variants with per-puzzle aggregation

Per-puzzle evaluation (All batches, ~1280 samples)...

Variant	Pass@1	Exact	Pass@2	Exact	Cell Acc
ms/puzzle N puzzles					
FP16	0.6850		0.6850		7.67
1000					
INT8 (bnb)	0.6910		0.6910		6.40
1000					
INT4 (fake)	0.0530		0.0530		3.19
1000					
FP32 (bf16)	0.6910		0.6910		3.09
1000					
INT8-emb	0.6980		0.6980		3.09
1000					

11.1 Per-channel calibration statistics

Calibrated INT4 helpers defined.

11.2 Apply calibrated INT4 and evaluate

12.1 QAT fine-tuning loop

13.1 Magnitude-based row pruning

Prune ratio	Actual sparsity	Puzzle	Exact	Cell Acc
0%	0.00%	0.6910	0.8747	
25%	24.96%	0.0000	0.5001	
50%	49.92%	0.0000	0.3781	

14.1 TorchScript trace of single TRM backbone step

Model: seq=81, emb_len=16, hidden=512, emb_dim=512

Trace done in 1.0s

Saved: trm_backbone_step.pt (19.34 MB)

Round-trip max abs diff: 0.00e+00 (expect ~0)

Edge deployment flow:

2. Load puzzle_emb.weights[puzzle_id] row (2.0 KB) from flash
3. Run traced(x, emb_row, z_H, z_L) for n_sup_max steps

14.2 Latency benchmark: traced model vs eager

Latency benchmark (20 runs, device=cuda):

Eager (float32) : 8.54 ms/step

Full inference (16 H-cycle steps):

Eager : 136.6 ms/puzzle

Traced : 75.8 ms/puzzle

File sizes:

trm_backbone_step.pt : 19.34 MB (backbone, FP32)

puzzle_emb.weights (INT8, full table): 0.5 KB

puzzle_emb row (1 puzzle, FP32) : 2.0 KB

15.1 Build train / val loaders

Train split : 800 samples

Val split : 200 samples

15.2 QAT loop with val-set evaluation every 10 steps

15.3 Plot train loss + val accuracy curve

[SKIP] No QAT history – run §15.2 first.

16.1 Build fused INT8 + SinglePuzzle model

Fused model breakdown:

INT8 backbone .pt size : 5860.4 KB (5.72 MB)

INT8 emb table (flash) : 0.5 KB (0.00 MB)

Active emb row (SRAM) : 2.00 KB

✓ SRAM during inference: 5862.4 KB (5.73 MB)

Fits 4MB SRAM? $\times$ Fits 8MB SRAM? $\checkmark$

16.2 Accuracy of fused model (val set, per-puzzle aggregation)

Evaluating fused INT8+SinglePuzzle model on val set...

Variant	Puzzle Exact	Cell Acc	ms/puzzle
FP32 (bf16) baseline	0.8700	0.9462	13.63
INT8 + Single-Puzzle (fused)	0.8600	0.9464	11.34
vs FP32	-0.0100 +	0.0002	

16.3 Save fused model (state dict + embedding table)

Saved state dict : trm_int8_singlepuzzle_state.pt (5.72 MB)

Saved emb table : puzzle_emb_fp32_table.pt (0.00 MB) $\leftarrow$ flash storage

Deployment package breakdown:

Backbone state dict : 5.72 MB (SRAM – INT8 weights)

Embedding table : 0.00 MB (flash)

Active emb row : 2.00 KB (SRAM per puzzle)

17.1 CPU-only latency benchmark

Moving fused model to CPU for edge-device latency simulation...

CPU latency (server CPU, FP32, 1 H-cycle step) : 36.7 ms/step

Full inference (16 steps) : 587 ms/puzzle

Estimated on-device scaling (order-of-magnitude):

Cortex-A55 (mobile, ~1 TFLOPS) : ~1.8s / puzzle

Cortex-M55 (MCU, ~128 GFLOPS) : ~11.7s / puzzle

ESP32-S3 LX7 (~40 GFLOPS) : ~29.3s / puzzle

17.2 Peak SRAM usage during inference (tracemalloc)

Peak SRAM delta during inference : 14 KB (0.01 MB)

Top allocations:

0.7 KB /lambda/nfs/EdgeTRM/TinyRecursiveModels/trm.py:212

0.5 KB /lambda/nfs/EdgeTRM/TinyRecursiveModels/models/layers.py:60

0.4 KB /lambda/nfs/EdgeTRM/TinyRecursiveModels/trm.py:114

17.3 FLOP estimate via torch.profiler

Estimated FLOPs per H-cycle step : 25660.1 MFLOPs

Full inference (10 steps) : 256.60 GFLOPs

Power estimate (Cortex-M55 @ ~1 GFLOPS/s, ~1 mW/GFLOP):
FLOPs : 256.601 GFLOPs/puzzle
Est. energy : ~256.60 mJ/puzzle
Puzzles per mAh (3.3V) : ~0

17.4 Deployment summary table

Backbone	MB	Total	deploy	MB	Puzzle	Exact	Cell	Acc	Fits	4MB	Fits	8MB
Config												
FP32 (baseline)					19.18			19.19		0.6910		
0.8747		x		x								
INT8 (bnb)					4.80			4.80		0.6910		
0.8751		x		✓								
INT4 (calibrated)					2.40			2.40		0.0530		
0.6602		✓		✓								
INT8 + Single-Puzzle ★					4.80			4.80		0.8600		
0.9464		x		✓								
recommended target for 8MB SRAM devices												

Utility: fix H_init / L_init device mismatch (permanent patch)

18.1 Evaluate all variants at n_sup_max = 1, 2, 4, 8, 16 (H_cycles = default vs 1) —

VARIANT: FP16		(device: cuda)					
H_cycles	Metric	n=1	n=2	n=4	n=8		
n=16							
3 (Def)	Exact	0.2830	0.4960	0.6100	0.6700		
0.7160							
Cell	0.7639	0.8151	0.8488	0.8677	0.8851		
Latency	0.8 ms	1.6 ms	3.1 ms	6.2 ms	12.3 ms		
GFLOPs	25.6	51.3	102.5	205.0	410.1		
1	Exact	0.0060	0.0210	0.3830	0.5480		
0.6390							
Cell	0.5526	0.6941	0.7856	0.8303	0.8574		
Latency	0.3 ms	0.6 ms	1.1 ms	2.1 ms	4.2 ms		
GFLOPs	8.5	17.1	34.2	68.4	136.7		
VARIANT: INT8 (bnb)		(device: cuda)					
H_cycles	Metric	n=1	n=2	n=4	n=8		
n=16							
3 (Def)	Exact	0.2800	0.5010	0.6210	0.6790		
0.7120							
Cell	0.7643	0.8183	0.8524	0.8721	0.8816		
Latency	0.7 ms	1.3 ms	2.6 ms	5.2 ms	10.2 ms		
GFLOPs	25.6	51.3	102.5	205.0	410.1		
1	Exact	0.0060	0.0210	0.3820	0.5540		
0.6560							

Cell	0.5528	0.6942	0.7852	0.8306	0.8638
Latency	0.3 ms	0.5 ms	0.9 ms	1.8 ms	3.5 ms
GFLOPs	8.5	17.1	34.2	68.4	136.7
VARIANT: INT4 (fake)		(device: cuda)			
H_cycles	Metric	n=1	n=2	n=4	n=8
n=16					
3 (Def)	Exact	0.0210	0.0290	0.0510	0.0550
0.0610					
Cell	0.6277	0.6490	0.6524	0.6573	0.6597
Latency	0.4 ms	0.7 ms	1.3 ms	2.6 ms	5.1 ms
GFLOPs	25.6	51.3	102.5	205.0	410.1
1	Exact	0.0000	0.0120	0.0210	0.0370
0.0510					
Cell	0.5612	0.6097	0.6375	0.6536	0.6555
Latency	0.2 ms	0.3 ms	0.5 ms	0.9 ms	1.7 ms
GFLOPs	8.5	17.1	34.2	68.4	136.7
VARIANT: FP32 (bf16)		(device: cuda)			
H_cycles	Metric	n=1	n=2	n=4	n=8
n=16					
3 (Def)	Exact	0.2840	0.4960	0.6040	0.6690
0.7090					
Cell	0.7640	0.8147	0.8474	0.8680	0.8810
Latency	0.8 ms	1.6 ms	3.2 ms	6.3 ms	12.7 ms
GFLOPs	25.6	51.3	102.5	205.0	410.1
1	Exact	0.0060	0.0210	0.3830	0.5470
0.6370					
Cell	0.5527	0.6943	0.7856	0.8293	0.8563
Latency	0.3 ms	0.6 ms	1.1 ms	2.2 ms	4.3 ms
GFLOPs	8.5	17.1	34.2	68.4	136.7

19.1 Zero-pad truncation at various context lengths

Full seq_len = 81 (FLOPs $\propto$ seq_len due to MLP architecture)

Context	% of full	MLP FLOPs	Puzzle Exact	Cell Acc
81	100%	~25.63 GFLOPs	0.2840	0.7640
40	49%	~14.80 GFLOPs	0.0000	0.3276
20	25%	~9.51 GFLOPs	0.0000	0.2035
16	20%	~8.46 GFLOPs	0.0000	0.1767

20.1 QAT from calibrated INT4 (lr=1e-6, 300 steps)

21.1 Instantiate student TRM and train with KL distillation

Teacher params : 5,028,866
 Student params : 744,962 (0.15 $\times$ teacher)
 Step 40/100 | kl=1802.0414 | hard=2.1362

Step 80/100 | kl=1125.4269 | hard=2.1942

Evaluating student on test set...

Student (hidden=256, 1L, 1-cycle): exact=0.0000 cell=0.5477 ~1
GFL0Ps

22.1 Monkey-patch softmax attention → linear attention

The Sudoku Extreme model uses the TRM-MLP architecture (mlp_t = True), which has no self-attention layers.

Linear attention approximation is not applicable. Skipping.

22.2 Retrain linear attention model to recover accuracy

No attention layers found in linear model. Skipping fine-tuning.
